

WHAT IS TOKEN ENGINEERING?

A stakeholder study

Funded through the Token Engineering Commons

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"We all know bridges require engineering. We all know they require care. We all know that other factors are involved in the construction. But at the end of the day, we all know we've driven over a bridge even if we don't know how it works. We trust it. We trust the people we will never meet".

Participant 30

Foreword: Building Future Infrastructure

Nascent. Rapidly evolving. Consequential. The field of token engineering represents an emergent space responsible for creating and connecting the invisible paths that humans, machines, and AI use to interchange, conduct commerce, communicate, and coordinate.

Since the creation of Bitcoin in 2009, we have seen significant developments in the realm of cryptocurrencies. The launch of Ethereum in 2015 created an explosion of crypto experiments. Before the introduction of Ethereum's smart contracts, the design space for token-based economies on production blockchains was quite limited. The term "token engineering" itself was not coined until 2018 after those crypto experiments grew mature enough and patterns started to form in the ecosystem.

The landscape of this nascent field was initially shaped by elements like blockchain infrastructure, governance mechanisms, privacy solutions, decentralized finance (DeFi), and multiple subsets of these topics. The permissionless nature of web3, rooted in open-source principles, grew to a level of complexity that prompted the development of social graphs to address challenges such as coordination. These social infrastructures incorporate reputation and reward systems, metrics for evaluating stakeholder contributions, and identity solutions to maintain a degree of privacy while preventing attacks. **Token engineering therefore functions as an orchestrator of the multidisciplinary facets of web3, with the common denominator of these elements being the creative use of tokens.**

In recent years, numerous token-based systems have been constructed, enabled by advances in blockchain technology. The decentralization ethos has reached many organizations facilitating cross-context collaboration. Efforts centered in regenerative economies matured and helped to funnel attention to public goods funding. Decentralized Science (DeSci) is evolving academic infrastructure, cooperatives-based models were brought to web3 and overall networked algorithms have contributed to operational efficiencies, deeper transparency, and a degree of complexity in how interactions are made in a multitude of successful projects. These are just a few of the benefits that continue to entice more people to build web3 solutions. And yet, while many positive examples exist, the broader web3 community has also experienced unprecedented disruption and reputational damage brought on by the failings of centralized exchanges, like FTX, the downfall of token experiments, like LUNA, failures due to smart contract vulnerability such as The DAO, bridge hacks, phishing scams, and what can only be described as either a lack of maturity, due diligence or a proliferation of pump-and-dump and ponzi schemes.

Despite these headline-grabbing setbacks and the regulatory uncertainty of the crypto market, innovation continues. Individuals and institutions within the blockchain sector are increasingly demanding common language and frameworks and desiring credible approaches for advancing their work.

Tokens serve as tools for decentralizing ownership, asserting social rights, and enhancing coordination strategies. These represent the ambitious goals that token systems aspire to achieve, along with the increasing consumer demand for greater privacy and autonomy.

While the overarching concept of token engineering may appear clear at a high level, the intricacies of its practice and the precise definition of its terminology were abstract and have inspired the undertaking of this study. Engaging with token engineering practitioners attempts to answer foundational questions to bring greater rigor and legitimacy to how token systems are contemplated, researched, designed, implemented, validated, and maintained.

Therefore, the primary question of this research study is: ***What is token engineering?***

In this report, we explore this question through the perspective of the people working within the field – their practices, challenges, and needs. We define the evolving boundaries that make up this nascent field, share the participants' thoughts on the future, and provide a discussion guide for further exploration of these topics.

We hope these findings highlight the evolving edges of the field and inspire continued experimentation and imagination around how it can shape our future world.

Livia, Lisa, and Nathalia

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How to Use

The research contained within this report was the result of a Token Engineering Commons Study conducted between March 2023 and March 2024, which was proposed in February 2023 and approved in March 2023.

This primary research represents the 41 voices who contributed to this study. It is meant to lay foundational work for further research and practice. The report is organized to help you learn what is most relevant to you regarding the token engineering field and its evolving nature.

Please reference the Table of Contents to jump to a topic of interest or simply go through it chronologically to get comprehensive insights.

As you move through the report you'll notice numerous participant quotes *in italics*. This report attempts to champion their voices and the insights they shared about the token engineering field.

In an effort to synthesize larger themes, summaries are shared at the end of key sections from the findings. These are meant to serve those looking to quickly understand key insights.

While the interview data and findings are the deliverables of this primary research report, and therefore we do not integrate external resources throughout, we have included a Discussion Guide and Recommended Resources at the bottom. These sections represent questions that were either shared by participants or sourced from researcher memos while analyzing the complete dataset, along with some recommended resources from our participants. The intent of sharing these questions and resources is to support the ongoing reflection of those invested in advancing this space.

Enjoy!

Attributions & Acknowledgments

This research would not have been possible without the generous backing of the Token Engineering Commons community and our team of professionals conducting and advising on the formation and execution of this study. A special thanks to **Danilo L. Bernardineli, Gideon Rosenblatt, Ataberk Casur, Rohan Mehta, and Shawn Anderson**, the reviewers and editors who helped in the finalization of this report.

Research Team

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Member of the TEC Coordination Team

Bear: General support.
Member of the TEC Coordination Team

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Study Participants

Our biggest thanks go to the 41 participants who shared their experiences and insights.

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Rohan Mehta
Simon de la Rouviere
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Trent McConaghy
Wassim Alsindi
Willy Ogorzaly

Participants were affiliated with diverse organizations, including but not limited to:

BlockScience
Common Stack
Common
DAOstack
Economics Design
Eight Arms Nine Brains
European University Institute
Fox Foundation
Hats Finance
General Magic
Giveth
Giza

Inverter Network
Kernel
Longtail Finance
MIT Computational Law Report
Ocean Protocol
Team Omega Solidity Audits
TE Academy
The ALANA Project
Token Engineering Commons
Token Kitchen
TokenGnomes
Untitled Frontier

Study Overview

Need for the Study

The rapid emergence and evolution of the token engineering field underscores the timeliness of this study. The application of token engineering permeates industries and societies. Yet, the opportunities for innovation are counterbalanced by technical challenges and ethical implications. Just like we depend on bridges, roads, and walkways to get us where we need to go in the physical world, token engineering is what can enable and incentivize our behaviors in the digital world - **the decisions made in this field today lay a foundation for the future world infrastructure.**

Token engineers grapple with the task of blueprinting the fundamental infrastructure of blockchain-based systems, leading to novel forms of value creation, exchange, and governance. Unfortunately, **this task is made more difficult with most people in the field lacking a line of sight to what peers or institutions are doing** to perform their work and advance the space.

The accelerated pace and demand can lead to a challenging work environment that expects products to “ship yesterday” in a field that has dire consequences and potential for harm if conducted poorly. Emergent and exponential technologies also pose the threat of unanticipated abuse and misuse of token-based systems, which requires deeper inclusion of diverse perspectives and deeper care throughout the solution creation process to “get it right”. **Token engineering holds the potential to bring a level of trustworthiness to a space plagued by doubt and cautionary tales.** While bridges collapse in real life, the public has ultimately come to trust and expect that they won’t. In the end, establishing that degree of public trust will be the ultimate measure of success in this new field of token engineering.

Purpose of the Study

This study’s primary purpose is to provide a tangible understanding of what token engineering is, its process, and its aims. This pioneering research will serve as an important milestone for describing a detailed exploration of the practices, challenges, and needs of those who are instrumental in shaping this emerging discipline as well as a historical account written at an inflection point in the development of these exponential technologies. As token engineering transcends conventional boundaries, fusing technology, economics, and ethics, this inquiry aspires to create a theoretical framework that is both academically robust and practically applicable. The purpose is twofold:

1. Understanding the Current State: The study seeks to capture the present state of the token engineering field, documenting the strategies and methodologies that practitioners employ in blueprinting the digital adaptive systems of tomorrow. By comprehending the current landscape, the research endeavors to unearth patterns and showcase key aspects of the token engineering practice.

2. Envisioning the Future: Understanding the present alone is insufficient. This research intends to lay the foundations for future developments by exploring the challenges and needs articulated by those within the field. By fostering a nuanced understanding of what is lacking and what is desired, the study aims to offer a beacon that can guide future endeavors in the token engineering arena.

The aim is not to provide definitive answers, but to illuminate the complexities and opportunities inherent in this domain.

Methodology

Study Design

This study was a collaborative effort, proposed to the Token Engineering Commons and approved via community governance. The study was designed with deep consideration for methodological congruence, aligning with web3 principles like data transparency and decentralized working, in addition to adhering to qualitative research practices. **The project team was assembled with no principal investigator, rather team members were enlisted with distributed roles according to background and expertise** aligned with their contributions to study outcomes.

Given the field's early formation, we chose to conduct a comprehensive and procedurally rigorous study, sharing findings in a full-length report, in hopes of creating a solid knowledge base to support subsequent studies and efforts within the practice of token engineering.

The study is rooted in a qualitative constructivist grounded theory approach. This approach emphasizes collecting data through in-depth interviews, making researcher memos, constantly comparing and analyzing data throughout the process, and producing insights others can challenge and build upon. Ultimately, this approach was selected because a) it is well-suited to exploring new or emergent phenomena, and b) it focuses on producing an overview of what is happening and how grounded it is in the data.

The interview questions were formulated with both academic and practitioner input. The semi-structured interview question set consisted of 14 questions, with additional prompts for elaboration or gathering further information as needed.

Participant Criteria and Recruitment

Participants were recruited based on their ability to adequately explore the primary research question, *what is token engineering?* This meant recruiting individuals who had past experiences in or were currently working within the token engineering field, rather than exclusively identifying as a token engineer by title, which is still an evolving role. The dataset is primarily represented by participants active in the Ethereum ecosystem. This serves as a limitation to our dataset, which would potentially produce different outcomes with the inclusion of a broader representation of other blockchain ecosystems.

Recruitment was purposive and snowball sampling, with the particular intention of maximizing the diversity of profiles being selected. We started with more prominent figures within the field and connections through the Token Engineering Commons for our initial participants and then asked the initial participants for recommendations regarding who they admired within the field and thought would contribute well to the focus of the study.

All participants in this study have conducted work within the broader token engineering field in recent years and represent roles of designing, implementing, auditing, researching, or offering expertise contributing to the launch of token-based systems.

Data Collection

To obtain a global view of how the token engineering field is emerging and evolving, 41 participants were interviewed between May and June 2023 by a team of three social researchers. Qualitative, semi-structured, intensive interviews were conducted on Zoom for 45-60 minutes per participant, using otter.ai for recording and transcription. Each interview had a primary interviewer, a memo-taker, and a secondary memo-taker, often listening to the interview recording at a later time and taking down additional notes. The interviews were auto-transcribed in *otter.ai* with members of the research team listening to the audio and verifying the transcriptions were recorded accurately. This often included minor corrections to the transcriptions and marking information that may put anonymity at risk, such as references to specific places or people.

Data Analysis

The dataset was analyzed by the three social researchers using constant comparative analysis, to develop categories and themes from the data. This approach consists of constantly reflecting on the data from the interviews, memos, researcher codes, and any other data gathered, such as suggested readings from our participants. Initially, thematic coding where key

concepts and researcher memos were documented, then followed by segment coding, which identified gerunds (action words from the interviews like *modeling* and *auditing* to extract what actions are taking place) and in vivo codes (keywords and phrases from the participants like referring to tokens as “the incentive of last resort”). **These codes were then used in weekly analysis discussions over two months for the researchers to align on key codes and themes from the data. The questions that guided these discussions were anchored in the data to make sense of “what is happening here?” and “what are the participants telling us?”** as they relate to token engineering, with particular attention to the practices, challenges, and needs of people working within the field.

The characteristics of each theme, their relationships with other data, contextual factors, and conditions contributing to token engineering's existence, as well as the consequences of these themes, were integrated to develop a theoretical framework. This framework offers a conceptual map and is constantly evolving, providing a dynamic depiction of the phenomenon under study. It aims to make sense of the complexity and nuances of token engineering, while also serving as a starting point for further research and practice.

Data Synthesis

In the formulation of writing this findings report, **the writing started with a chronological view of the dataset, by consolidating the data over each interview question.** Each question's dataset was combed for common codes and themes, anomalies, tensions or contradictions, and key ideas that generated a lot of energy and questions from the participants. This guided what would be written and surfaced in the report – to give a pulse of how the group of 41 participants was reflecting on, contesting, and questioning a given topic.

After drafting a consolidation of data for each interview question, we examined the dataset for outliers and themes that were not directly related to the specific themes of the questions but emerged across responses or were raised by participants unprompted. To address these themes new sections of the report were created like the Process section, which was reflected on by nearly every participant, yet in different ways and at different points within the interviews.

The intent with the inclusion of any new section was to assist in describing patterns that emerged, which would shed light on the practices, challenges, or needs of these participants. The final sweep of data was to look at common questions raised and resources recommended to share with those interested in continued reflection on the insights generated from the study.

Finally, we organized and interpreted the synthesized results such that the report structure and content would conform to the following outcomes:

- 1) share the study overview and context for the dataset
- 2) describe the process and practices happening within the field

- 3) represent the challenges and needs shared
- 4) articulate participants' hopes for and perspectives on the future
- 5) provide a discussion guide and resources for further inquiry

The themes presented were refined and revised through ongoing data collection and analysis. External reviews and feedback were solicited to identify any gaps or opportunities to further clarify the findings. Intentional efforts were also made to follow up with participants, or subject matter experts, regarding emergent data or specific inquiries to ensure we were accurately representing the knowledgeable perspectives of our participants.

Data Storage

The full dataset of this study, comprising 41 full interview transcripts, has been cleaned to protect anonymity and includes researcher codes. It is available in persistent storage hosted by the Token Engineering Commons. The data is open, accessible, and available for anyone interested in exploring, replicating, challenging, or expanding upon the research.

Participant Overview

The findings shared in this report represent the contributions of our diverse set of participants. Each contributed distinct perspectives on the field with varying backgrounds. To protect the identities of participants that opted to remain anonymous, we refer to them using a numbering system #1-41.

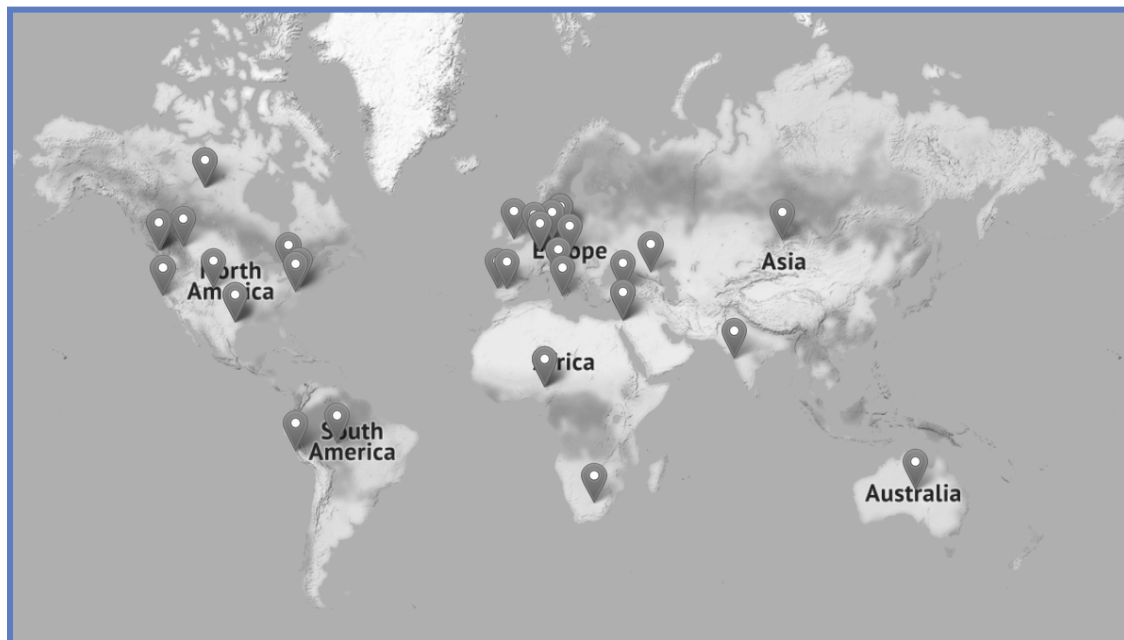


Fig. [ParticipantLocations]

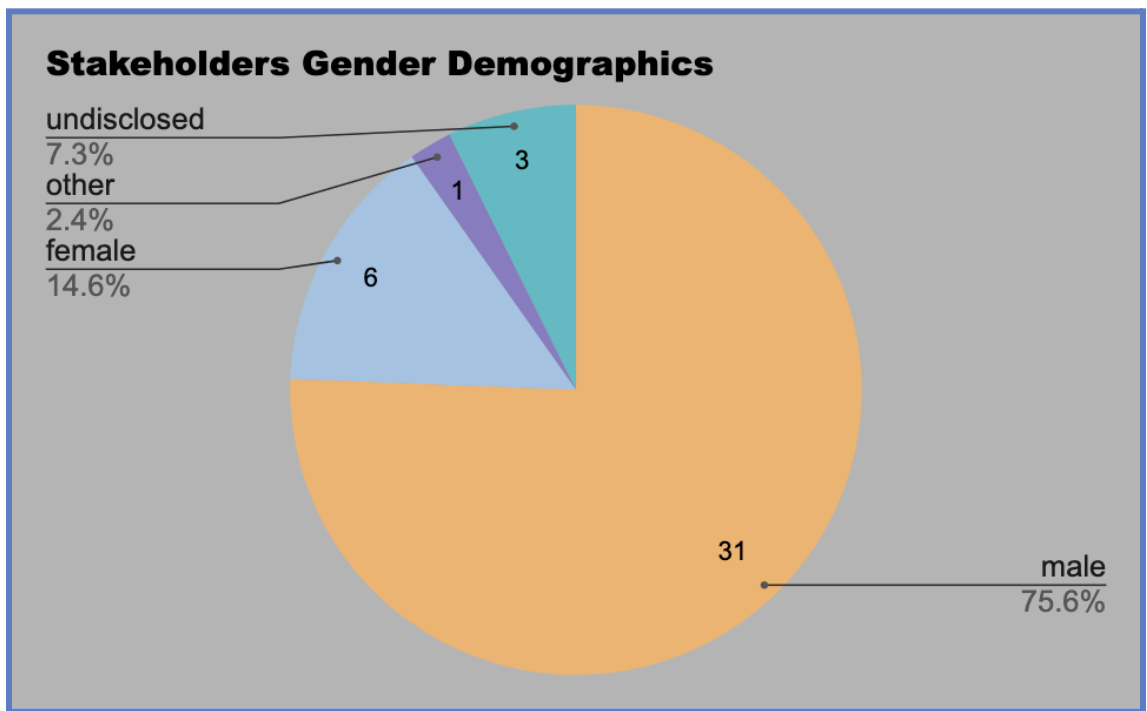


Fig. [GenderDemographics]

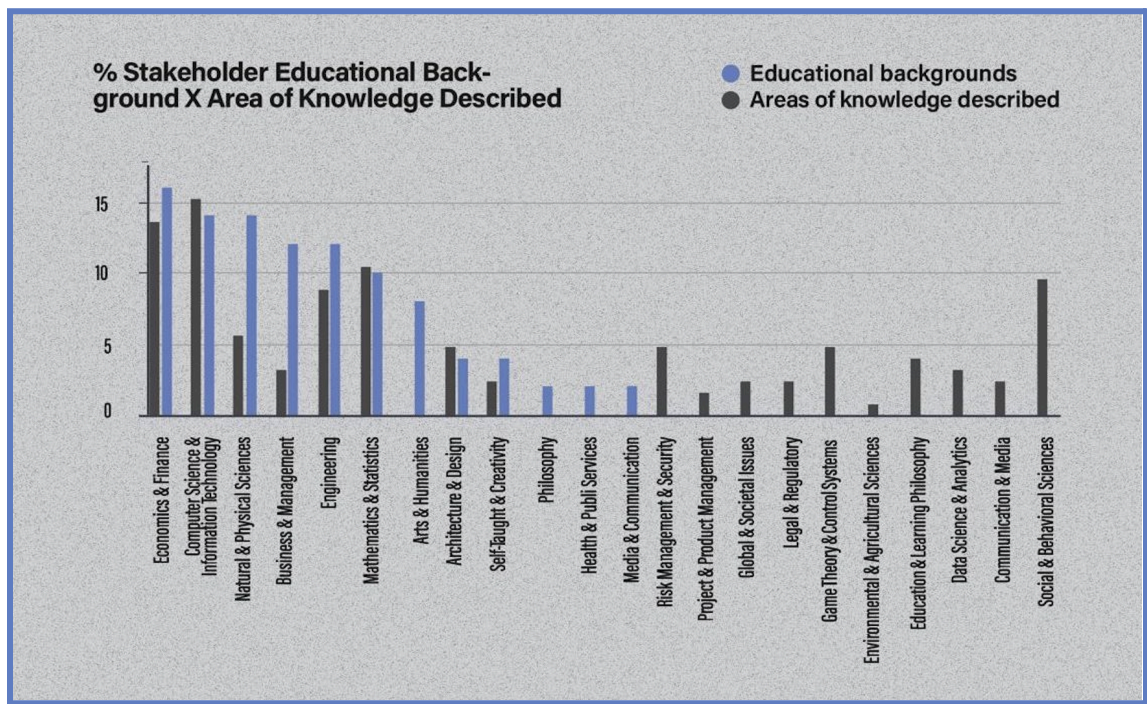


Fig. [EducationalBackgroundxAreasOfKnowledge]

Participants had predominantly STEM backgrounds with a variety of educational influences, showcasing multidisciplinary approaches. The participant group consists of a male majority with most of the participants based in countries associated with Western/European identities. There is a relevant number of them located in the Global South. Interestingly, the number of participants in the metropolitan/developed areas of the Global South and non-Western areas is disproportionately rare.

Participant Profiles

The participants showcased a wide breadth of backgrounds, encompassing both technical and non-technical disciplines. From those immersed in start-up culture to researchers with deep scientific expertise, the diversity of experiences laid a robust foundation for exploring the field. Their motivations for entering the world of token engineering were equally diverse, ranging from an interest in the liberating potential of the space to redefining economic principles in the context of web3. Some participants joined the field due to their curiosity about blockchain's impact on public health and environmentalism, while others aimed to leverage token engineering for social good and the advancement of decentralized technologies. The majority of participants represented Ethereum as the main Layer 1 (L1), while a few others represented chains like Celo and Filecoin. Their collective experiences and insights underscore the interdisciplinary nature of token engineering and its potential.

Participant 1: Their entrance into cryptocurrency began in 2015-2016, initially enticed by the concept of the "internet of money." With a background in computer science, statistics, finance, and game theory, they delved into blockchain technology, exploring smart contracts, blockchain virtual machines, peer-to-peer networking, and system design. They currently lead a renowned economics R&D team.

Participant 2: Their entrepreneurial journey began during college when they transformed a class project into a successful startup. In 2017, their interest in cryptocurrency bloomed, leading to their role as a product manager in a prominent DAO in 2018, where they contributed to designing and launching DAOs.

Participant 3: With a background in high-energy physics they transitioned to economics, exploring simulations and agent-based modeling. This path led them to perceive token economics as a means to better understand and engineer agreements within the blockchain realm for tangible use cases. Their work delves into community-inclusive currencies and mutual credit systems.

Participant 4, Wassim Alsindi: With a diverse background spanning physical sciences, arts, commerce, finance, and philosophy, their cryptocurrency journey commenced in 2013, driven by blockchain's liberatory potential. They conducted research blending computer science, data

analysis, and epistemology, ultimately founding a publication focused on governance, regulation, and decentralized systems.

Participant 5, Trent McConaghy: Having nurtured an early interest in programming, they studied computer science & engineering, and established AI / chip design companies focusing on CAD & simulation. They obtained a PhD in engineering doing AI, and became involved in blockchain and token technology in 2011. They drive a top blockchain/AI project, do applied token engineering, and write about it. Their work continues to shape conversations in the field.

Participant 6, Success D. Ndu: With an economics-focused background, they initially traded agricultural commodities before transitioning to full-time cryptocurrency involvement in 2020. In 2021, they delved deeper into the field, aiming to advance token engineering's role as a sustainable career path.

Participant 7, Stella Achenbach: Their background lies in problem-solving through design, with early influences from experiences in 3D and digital creation leading them to the token engineering field. While the term "token engineering" may not fully encompass their focus, they apply design theories to web3 challenges.

Participant 8, Simon de la Rouviere: Their cryptocurrency journey began in 2013, intrigued by the idea of cryptocurrency supporting various endeavors. They have contributed notable innovations to the space and focus on building tools to support creators in the blockchain space, aiming to empower creators with new technologies and storytelling.

Participant 9, Shermin Voshmgir: Shermin has been involved in the earliest conversations shaping the field and has written various books on the token economy and how to design and analyze token systems. She also developed a holistic design thinking tool for Token Engineering which covers all aspects of the engineering process - technical, economic, legal, and political-ethical dimensions.

Participant 10, Shawn Anderson: Joined the token engineering field as a data scientist, offering services to the industry. Their resources and contributions reflect an open-access approach. Their active involvement in collaborating and educating others began in early 2020.

Participant 11: Transitioning from pursuing a Ph.D. to becoming a software engineer actively involved in DAOs, they explored concepts like Conviction Voting and DAO frameworks, ultimately focusing on developing tools for simplifying tokenomics systems for DAOs.

Participant 12: A computer scientist with expertise in the automation of agent systems, they delved into decentralized work, founding an organization in 2016. Their journey into token engineering blends a pragmatic approach with embracing new paradigms of thinking. Their work exemplifies the exploration of tokens' roles in coordinating human and machine systems to impact real-world outcomes.

Participant 13, Rohan Mehta: With a master's degree in Physics and a BE in Electronics and instrumentation engineering, they pursued a career in data science. Disheartened by the challenges of obtaining clean data and deriving actionable insights, they pivoted towards the field of token engineering, particularly attracted to socio-technical systems. They are now involved in building AI bots for token engineering standardization.

Participant 14: Studied economics with a focus on labor and environmental aspects, delving into Chinese economic development. Founded a startup in rainwater harvesting and tokenized it during the early days of blockchain. Currently explores token engineering, emphasizing the transition from digital commons to physical commons, and is creating a circular food, energy, and fiber marketplace using land as an "earth bank."

Participant 15: Data scientist and business strategy professional with a background in economics and mathematics. They discovered token engineering through the TEC and the TE Academy and contributed to the early stages of token development and design while currently pursuing a master's program. They are actively engaging in communications and research promotion related to token engineering.

Participant 16, Pedro Parrachia: Their diverse career encompassed public health, environmentalism, and human rights, eventually leading them to blockchain in 2016. They are currently at the forefront of digital-native regeneration, designing systems where web3 technologies and ecological principles converge to create sustainable futures. They have strategized and implemented dozens of relevant projects over the last few years.

Participant 17, Octopus: Their journey began with learning calculus, leading to a Ph.D. in mathematics and a career as a professor, honored with multiple collegiate teaching awards. In 2020, they rekindled their interest in cryptocurrency, contributing to data analysis and research and partnering with pioneering organizations in the field.

Participant 18: A journey spanning technology, AI, blockchain, and finance led them to explore distributed ledger's immutability and open finance, as part of their emphasis on the need for financial literacy within the blockchain ecosystem.

Participant 19, Michael Zargham: With a background in large-scale, multi-agent control systems engineering, they maintain ethical practices at the forefront while advancing the token engineering field, drawing attention to the ethics and design practices in emerging public infrastructures and web3. Their work within the field is among the most admired and has deeply influenced the industry.

Participant 20, Matan Field: Holds a PhD in theoretical physics, specializing in unification theories. Concurrently, they developed an interest in cooperatives, founding an organic food cooperative. Explored decentralized ride-sharing applications, pioneering token engineering through a unique proof-of-movement concept. Co-founded a company focusing on technology protocols for DAO governance.

Participant 21: Transitioned from electrical engineering to entrepreneurship and ultimately discovered blockchain technology. They were present at the birth of ERC-20 and ERC-721 token standards and played an active role in the emerging NFT space.

Participant 22: Background in economics and tech startup expansion. Entered web3 with skepticism and recognized the need to rebuild the fundamentals in the context of blockchain technology. Spent two years developing a conceptual framework for token engineering and is focused on creating new incentive structures within web3.

Participant 23: Holds a master's in mathematics and concluded a PhD focused in crypto-economics. They are dedicated to scientifically and academically founded methodologies for designing token systems. Despite their academic background, their contributions to the field prioritize the advancement of protocol design with a multidisciplinary emphasis over academic titles.

Participant 24, Kris Renzo: In 2018, they transitioned from architecture to cryptocurrency research. Their work at an online database for the crypto industry fueled their interest in token economics and led them to discover token engineering. They have a continuous focus on building their skills and knowledge alongside producing meaningful contributions to the field.

Participant 25: Started exploring cryptocurrency in 2014 through a play-to-earn game. They eventually pursued a degree in economics and have been contributing to systems, incentives, and statistical design since 2018. Their focus is to help web3 gaming companies design sustainable economies.

Participant 26, Kelsie Nabben: Dr Kelsie Nabben is an ethnographic researcher specializing in the social impacts of emerging technologies, notably decentralized digital infrastructure, and Artificial Intelligence. Her research focuses on resilience and accountability in contexts of digital, self-governance. Kelsie is affiliated with BlockScience as a researcher, European University Institute as a Max Weber Fellow, BlockchainGov, and RMIT University.

Participant 27, Jessica Zartler: With a diverse background spanning journalism and political science, they have been leading communications for a range of web3 projects over the last years. Their work is focused on communications and research at BlockScience, they are a course author and advisor at Token Engineering Academy, and a founding steward of the Token Engineering Commons.

Participant 28, Jelle Gerbrandy: With a Ph.D. in logic, they moved into software development, and found they could combine these two experiences in the blockchain space, where they worked as a developer, as Chief Technology Officer (CTO), and as a token engineer. They have deep experience with audits and the often-overlooked back-end part of the token engineering process.

Participant 29, Jeff Emmett: With a background in electrical engineering, they were drawn into the field by the potential of web3 tools and a perceived gap in communication that excessively emphasized the 'how' instead of the 'why.' They focus on the composability and interoperability of token system primitives, leveraging their prior knowledge in circuit design.

Participant 30: Their background in economics and mathematics is focused on incentive structures and mechanisms to motivate rational actions. They transitioned from academia to consulting and embraced blockchain and smart contracts.

Participant 31: Coming from a background in game development and online advertising, they co-wrote a paper on token-curated registries, contributed to a significant DAO's white paper, facilitated the launch of the first payment channel on the Ethereum Mainnet, and played a crucial role as an incentive designer and technical coordinator. Currently, their focus is on decentralized IDs and account abstraction.

Participant 32, Griff Green: With a master's degree in digital currency, their interest in token engineering began with The DAO in 2016, influencing their dedication to innovating and improving systems beyond traditional government structures.

Participant 33, David Sisson: Combining an academic background including a PhD in Neuroscience with commercial experience developing automation and data systems for science, engineering, and business, they now work on evolving the conceptual aspects of the token engineering field, ensuring a comprehensive understanding of the token's operating context and their role in the complex systems built on peer-to-peer networks.

Participant 34, Danilo Lessa Bernardineli: With an interdisciplinary background and approach, they started by replicating articles related to public policy and health using modeling tools. They were working as a data scientist before immersing more deeply into the token engineering space and their modeling work led to their involvement with various projects in modeling complex systems.

Participant 35: Focused on bringing together entrepreneurs to address global challenges, they turned their attention to token engineering in late 2022. Deeply interested in building sustainable token economies, they explore new incentive structures and value redefinition within the blockchain space.

Participant 36, Andy Tudhope: With a background in English literature and fascinated by the cultural diversity of blockchain users, they got involved with lawyers, Swedish anarchists, and holocracy enthusiasts before the launch of Ethereum. They co-founded an early protocol-level infrastructure and consumer applications project and aim to create value-driven solutions in blockchain technology.

Participant 37, Cem Dagdelen: With a background rooted in economics and finance, they transitioned from business strategy and management consulting to mechanism design and

token engineering. They emphasize their affinity for creative, design-oriented approaches within the field.

Participant 38: Pursued data analytics in January 2020. Their interest in blockchain technology began in 2017, initially driven by financial gain and transitioning into token engineering in January 2021.

Participant 39: Protocol and mechanism designer at a prestigious institution with a PhD in Algorithmic Game Theory.

Participant 40, Ataberk Casur: Deeply interested in the symbolic nature of tokens and their role in representing values within specific systems or communities, they co-founded an organization and engaged in protocol architecture and tokenomics.

Participant 41, Angela Kreitenweis: With a background in design and early-stage venture development in web2, they are interested in the potential of tokens as rewards for value-add and coordination. They explore human behavior within systems and their contributions have been shaping token engineering education.

Defining Token Engineering

To answer the essential question of this study – “What is token engineering?” – it is useful to define each of these words:

Token

Tokens were often referred to by participants as *a store of value* or an asset on the blockchain. One interviewee highlighted that “*tokens are cryptographic abstractions of information and/or value that can flow around the systems like atoms*”. In the context of chemistry and physics, an atom is the fundamental unit of matter and the basic building block of all chemical elements, just as a token is a building block of blockchain-based value exchange systems.

Our participants shared various perspectives. Some saw tokens as a representation of information and as a means to engineer social systems. Others viewed tokens as a tool for novel financial experimentation. **Tokens can hold one or multiple functions of value simultaneously and can be programmed as credit, reputation, vouchers, shares, and rights.** They were described as a medium to optimize objectives, enabling new forms of value exchange, ownership, and accreditation representation. Their programmability and trackability are at the core of token engineering.

Engineering

Why an emphasis on *engineering* compared to terms like *token economics*, *tokenomics*, *token design*, and *mechanism design*?

Two tenured participants referenced Wikipedia’s definition of engineering as a relevant starting point. At the moment of this study, one of the relevant portions of that definition was the following:

“The American Engineers’ Council for Professional Development¹ (2022- 2023) has defined “engineering” as

The creative application of scientific principles to design or develop structures, machines, apparatus, or manufacturing processes, or works utilizing them singly or in combination; or to construct or operate the same with full cognizance of their design; or to forecast their behavior under specific operating conditions; all as respects an intended function, economics of operation and safety to life and property. “

Between these two participants, one highlighted that design is present within engineering, while the other reiterated that it's a process of **design, verification, deployment, and maintenance** of artifacts. Using the term engineering acknowledges the whole process. Speaking about *token design* alone may thus understate the importance of these other practices, which are directly related to the ethics and safety of what is being created.

Safety was a predominant word used to describe the relevance of engineering. A participant shared, “You design for and reasonably expect people to consume them [artifacts/public infrastructure] without having to individually or independently validate their safety or integrity.” When someone crosses a bridge, they don't need to test the safety of the bridge first, they assume this already happened in the engineering process otherwise the bridge wouldn't be there.” The term engineering seems to extend that same rigor to web3 systems.

Of course, the word *engineering* is not enough, by itself, to guarantee safety and ethics are being applied. It does, however, insinuate something like a Hippocratic Oath of *do no harm*, which many participants believed was missing or needed for the field to advance successfully.

As with the word *design*, many participants found terms like *economics* to be too limiting to the various uses and possibilities of tokens, which go beyond the decentralized finance or treasury allocation use cases that are often associated with using tokens within web3.

Summary: *Tokens* are referred to in this report as blockchain assets and *engineering* is a scientific approach to building these token-based systems with a degree of rigor, safety, and ethics.

The Role of a Token Engineer

Some participants expressed discomfort with the term *engineering* because, despite their knowledge and expertise, they don't have a formal engineering education.

Do I need to be an engineer to be a token engineer?

One may argue that someone **who can navigate the steps of design, implementation, and verification of web3 artifacts is a token engineer**, and others may argue that each of these phases imposes complexity in different subjects that are better suited for a team to address together. One participant put it this way, “it's too much for even one whole company, let alone an individual engineer.”

Many others expressed that **a natural progression for the field is toward greater specialization in each phase** of the process and that organizations should expect to hire groups and teams to accomplish the work versus looking for one person. This is especially true

when the product is particularly complex or has massive impact potential. Specific comparisons to other roles were also made by some participants. One participant reflected that the role of a token engineer is like a product manager, ushering the work through all the needed phases to produce a quality outcome. Another participant described the role as a system's auditor, capable of analyzing the interconnections of that particular complex system to attest to the overall safety and usability of what has been created. While there may not be an agreed-upon job description for a token engineer, nor established accreditation criteria for this role, there was major agreement on the importance of clearly and proactively identifying responsibilities for both clients and practitioners.

"I think it's a question of time. I don't think people have a good mental image of what token engineering is as a profession. I think that people are overly focused on "I want to be a token engineer", and "I want to do simulations" but they don't have this visibility that the token engineer is not only about the engineer but also about the analyst, about meetings, people talk, compliance, entrepreneurship. It's the entire ecosystem. **Most of the people who interact with that field are not engineers themselves.** And I don't think that people have a good grasp of that. Danilo Lessa Bernardineli

Summary: The job description of a token engineer remains ambiguous and seems to look more like a process that requires lots of roles and teams versus one position. The field may benefit from building upon well-established criteria and practices from the field of engineering.

The Token Engineering Process

“Birthing life to a digital soul.” Ataberk Casur

Is there a step-by-step process to token engineering?

When engineers build bridges or highways, they are not engaged in the construction work itself. Instead, **their responsibility lies in the blueprint – the well-laid plans utilized by others for the actual construction work serving the public.** They ensure that the construction execution aligns with the intended design and that its conditions remain continuously adequate for the intended usage.

Similarly, token engineers create the blueprint which they may develop or hand off to software developers or others to then construct the work. Creating a well-laid blueprint and ushering the work through various phases is the process of token engineering.

The participant responses resoundingly conveyed that an end-to-end process is not clear or established for most people working in the field, though they indicated that such clarity would be helpful. Although not all of our participants agreed there was a process to be articulated, key distinctions and phases emerged across the responses. This section outlines the five general phases of the token engineering process: **discovery, design, implementation, verification & validation, and maintenance.**

Phase	Discovery	Design	Implementation	Verification & Validation	Maintenance
Associated With	Initial scoping	Modeling	Translating theoretic models into computational representations	Feasibility check Safety check	Monitoring, analyzing and improving
How it fits within a prototypical product launch journey	Before Testnet	Just before Testnet, or between Testnet and Mainnet	Just before Testnet, or between Testnet and Mainnet. Can be fully concluded after Mainnet launch	Before Mainnet launch. Can be fully concluded after Mainnet launch	After Mainnet
Key Activities	Breakdown of requirements Collect data from similar systems Acknowledge assumptions Define a clear scope for the TE project	Define the goals of the system, KPI and metrics. Incentive design. Propose initial conditions and parameter sets. Make technical choices	Simulate, observe, iterate, and Adjust parameters.	Smart contract audits. Incentive structure verification. Token engineering audit.	Validation with stakeholders, Data and behavior analysis Suggesting improvements
Key Question	What are the needs of this project?	What behaviors should be incentivized to support the system goal?	Are the formal specifications representing the system's intended operation?	Is this system safe for use? What might transpire?	Was product-market fit achieved? Are the metrics useful?
Key Quote	"The models are only as good as the assumptions you have"	"The token is not the meal, it's the seasoning." David Sisson	"The more simulations that we can do, the less risk [there] is going to be to deploy tokens that may fail." Participant 11	"Do not trust but verify." Jelle Gerbrandy	"No plans survive contact with reality." cited by Participant 31

These phases are iterative and may not occur linearly. It is common for feedback loops to happen, in which the conclusion of a phase ties back to a previous one rather than the next. This can happen as a consequence of requirements changing, new information being revealed, and technical and regulatory choices requiring revision. Since these systems are made up of many elements, reiterating is part of the process.

Participants offered details and examples of the work they perform and how they ensure quality solutions. The following sections represent a further articulation of each of the phases of the token engineering process.

1. Discover

Description: The discovery phase can be described as the initial contact with the client and the breakdown of the project's needs. This is where conversations evolve, **transforming the problem-to-solve into a clearer scope**, which can have different outcomes (such as a written document, diagrams, or roadmaps) depending on the project. This is the phase where both clients and token engineers will determine the right engineering approach for the project and what expectations should be made over the future life-cycle of the engagement.

Challenges: In this phase, we heard challenges around the lack of clarity from stakeholders regarding their needs and the difficulty in communicating complex concepts and processes.

Highlights: A highlight of the discovery phase was the importance of identifying the assumptions being made at early stages as well as identifying and mapping as many unknown variables as possible.

Excerpts from the interviews:

“There's a major sort of presupposition there that the people asking the questions know what they want the token to do. They may have this broad conceptual idea of what they want to do, but it's going to be operating in a concrete context. And understanding, you know, the complexities of that context are critical to get the outcomes the people want to get. That's where I work.” David Sisson

“Be very, very clear, what is the incentive problem? What incentive structure is actually broken? Why do you want to fix it? Why are you well-placed to fix them? And only then might you begin a project” Andy Tudhope

“When do you have to have a token? When there's no other way to align incentives. It's the incentive of last resort.” Participant 31

2. Design

Description: The design phase is where requirements and desirables, as understood in the discovery phase, are refined. The goals of the system are established, and metrics for its evaluation are proposed. Agents' behaviors are examined in deeper detail, and incentive mechanisms are created to guide stakeholders' actions within the system. **This is the phase where technical choices are made regarding substrates, primitives, platforms, and tools to be used.** The design phase will identify and prepare the parameters and conditions for the simulations conducted in the implementation phase by initiating agent-based modeling and algorithmic design.

Challenges: There is often a reconciling of conflicting goals during this phase. Modeling the practitioner's decisions to be made about what the ultimate goal is and what aspects are more important than others. As exemplified by Participant 1, sometimes people agree on the long-term purpose of the economy but disagree on how to get there. The challenge in this phase is also to develop incentives for stakeholder engagement that reinforce the values and goals of the system without introducing unintended harmful consequences.

Highlights: Participants clarified the difference between creating new tokens and working with third-party token flows as part of possible actions stakeholders can perform in the system. David Sisson highlights that TE's main concern is the intertwining of mechanisms and stakeholders and not necessarily the creation of tokens. They add that having tokens as a primary focus can be a distraction depending on the context, but that doesn't mean that tokens aren't present at all.

Immutability is an important feature of blockchain technology, but certain aspects of a system are more mutable than others. When setting initial conditions, it is crucial to think about all steps and anticipate how particular components can be parameterized in the future without having to redesign the entire system. Governance was mentioned as a specific example to consider. The rapid evolution of governance mechanisms and the decision of certain projects to progressively decentralize their decision power could bring about many changes. Considering these potential changes before setting the initial conditions could contribute to the creation of more resilient systems.

Excerpts from the interviews

*"I think this is probably the most important message to the token engineering community now: **initial conditions matter**. You will never have a perfect system, but any flaws that you need to retroactively amend whenever upgrading rules can be hard because of existing power structures."* Shermin Voshmgir

"I think you should start from the question: what is the value that you want to create?"
"[step] Number one is what is the collective outcome you're looking for? Step number two is what is the individual action that you're seeking to incentivize that will trigger the collective phenomena value creation?" Matan Field

"The token is not the meal, it's the seasoning. It can bring out the flavors of food, but you can't be nourished by just eating salt. It can enhance what's there." Participant 31

3. Implement

Description: Computational implementation is a computer science term that implies that the original design of the system is faithfully preserved, with its essential properties and relationships, when converted from a model (such as a theoretical idea) to its computational representation (such as software). In this phase, the initial conditions and parameters set during the design phase are translated to formal specifications that can take the form of mathematical equations and algorithmic descriptions or pseudocodes to generate simulations. Simulations and models have the role of facilitating the understanding of the system being designed and can validate or break some assumptions so that changes and reiterations can be made as needed.

Challenges: Implementation may require attentive refinement of parameters and specifications to ensure the computational visualization represents the model and stakeholder goals. When the simulations are ready to be shared with the stakeholders, another challenge emerges, which is ensuring the content is legible and accessible to them.

Highlights: Implementation methods are not siloed in a linear order. They can be present in the design, verification, and maintenance phases since the **core of implementation is the translation from one model of representation to another, and the optimization aspects that come from observing and iterating on the system behavior**. Implementation methods described by our participants include:

- Algorithmic design: Involves defining the input, specifying the desired output, and formulating a series of well-defined steps that will transform the input into the desired output.
- Agent-Based Modeling and Simulating: Modeling and simulating complex systems by representing individual agents and their interactions with predefined rules for agent behavior and environmental conditions.

Excerpts from the interviews:

“Token engineering is how to optimize the objectives that we're looking at using tokens as a medium.” Participant 22

“I wrote down the objectives and constraints that I wanted, and then I treated it like an optimization problem. And then I started applying solvers, which was, in this case, applying different building blocks, different token designs that existed, and so on, trying different things, some worked, some didn't, I would analyze against my criteria. (...) so I was able to go from a state of failing to a state of actually coming up with a design that we went with (...) [this was] done in a way that we're proud of so I wrote about that, gave talks about it, and that helped to kick off token engineering.” Trent McConaghy

“It is the process of continuously learning and getting feedback and analysis back into it, such that you might have to change parameters or have some artifact come out of this process to learn for future implementation.” Simon de la Rouviere

“Mathematical specifications are a big part of that [referring to what they do]. Because you want to place that into a quantitative context where you can get as much predictive power as you can. That's often taken in the pipeline, and it will be at least a part of it at the beginning of a computational implementation.” Participant 30

4. Verify & Validate

Description: Verification and validation practices form the backbone of token engineering. **This is what differentiates simple design processes from an engineering discipline with the rigor to build relevant systems** that protect stakeholders from potential risks and vulnerabilities.

There was a fair amount of disagreement between participants regarding the use of the words verification and validation. Further review showed us that this is a common debate in the engineering field in general, so we opted for the most consistent examples within our dataset and described how we are using these words below.

The validation process ensures design choices are technically feasible and effectively meet the requirements and expectations of stakeholders, including considerations such as product-market fit. Validation focuses on alignment with user needs and external suitability. Another way of putting it is that Validation ensures that the outcome matches with the intent/goals.

The verification process examines the safety, integrity, and reliability of the system. This process involves rigorous audits of the code, incentive structure, and the system as a whole. In opposition to validation, Verification ensures that the outcome matches the description and/or specification.

Forms of **verification and validation** mentioned in this study include digital verification, analog verification, and token engineering audit.

Digital: smart contract audits.

Analog: incentive structure analysis.

System audit or token engineering audit: overall risk assessment, validation of the system's accuracy including procedures and rules for how the token should and should not be used in the protocol. Interoperability capacity, tokenomics review, system performance evaluation, product-market fit, and regulatory compliance.

Highlights:

Trent McConaghy mentioned, “We can take theory, practice, and tools from analog, digital, and mixed-signal verification in electrical engineering, and port them for verification in token engineering. Digital/logic components are discrete in time (clocked) and value (typically True/False). Analog components are continuous-time, or continuous-value, or both.

Mixed-signal components have both analog and digital.

*Digital verification is fairly well-developed in token engineering; the vast majority of audits and tools focus on it. But they miss analog elements such as incentive design, extreme values in flash loans, or race conditions in token bridges. **This is where analog verification in token engineering can help a lot, but we are still in the early days.** Analog token engineering simulators like cadCAD and TokenSPICE are the base-level starting point. There remains a great opportunity for more analog and mixed-signal verification tools, such as simulator-in-the-loop software to identify failing edge cases.”*

Griff Green uses the term “**token engineering audit**”, which we decided to adopt in this report to describe what other participants called system audit. They advocate for the existence of formal processes to audit token systems towards the development of safety standards specific to web3, which could include aspects like liquidity strategies, stakeholder engagement, and regulatory compliance.

Challenges: External pressures to “**ship yesterday**” often create tension to speed through due diligence, which has led to numerous bad outcomes in the market that have created reputational harm for the entire space. Going slow to go fast ensures these phases create the necessary pauses for the project because once it is on-chain and being used by multiple stakeholders it is much harder to change.

Not only code can go wrong, but also human actions that are led by designed incentives. This is the moment when the complexity of the system needs to be taken into careful consideration. Therefore, professionals with a good understanding of the entire process are best suited to perform a token engineering audit, although study respondents noted that these people are still scarce in the field.

Excerpts from the interviews:

“Generally this is a macro view that’s looking at an ongoing forecast under different market scenarios of the quantitative impact of all relevant value flows. Understanding every type of asset in the economy, whether it’s on or off-chain, fungible or non-fungible, we want to understand inflation, understand the impact of the different economic agents, and understand how they might change under various scenarios. So understanding what impact a certain type of player persona will have on the economy based on the different decisions they make... And then the info or the data derived from that simulation can also be used to do a more macro simulation of understanding if there’s a certain number of this type of user personas entering the

economy. What is the larger simulated impact on the system and whatever goals you're trying to figure out?" Participant 25

"System audits basically, and not like contract audit, but just be playing the devil's advocate for whatever design that you make." Ataberk Casur

"From there, we go to the validation which is like, how do I know the design is good, right? What are the metrics we care about" Participant 1

"I see the code, but sometimes it's not clear whether this code implements what [the audit's client] said it should do, and maybe it should not do that at all. Maybe [the written code] is just really a design fault. So what we do with auditing is also to go into that part. This is very much appreciated by our clients, and it's more fun work as well." Jelle Gerbrandy

End-to-end process:

Significantly important is the role of the verification and validation phase throughout the end-to-end process. Verifying and validating were mentioned as important after each of the other phases. This diligent work is really what makes token engineering an actual engineering discipline. There is a continuous emphasis and effort put toward ensuring that the desired outcome and actual outcome are congruent. Here are just some examples of how the verification and validation phase permeates many parts of the process:

1. Token engineers gather the requirements for the project in the discovery phase and **validate** the scope with stakeholders before designing models.
2. During the design phase, token engineers sketch a proposal for how the system should function using models and diagrams and **validate** the design with stakeholders and their internal team "Are we sure our proposal is implementable? What metrics do we care about? Is it practical?" Participant 1
3. In the implementation phase, token engineers translate the models into computational representations like simulations.
4. Code audits and other **verification** methods are used to ensure the system is safe. Then a token engineering audit determines if everything is well-connected and ready to be launched.
5. After launch, the system is monitored by periodic **validation and verification** processes to ensure it remains safe and that it is achieving its intended purpose.

Furthermore, Danilo Lessa Bernardineli highlighted that **the personal journey of a token engineer might be very different from the project journey** since TEs could be joining projects that already have a clear scope and a design they want to implement for example. So it is important to know the entire process, but it does not mean you or your team will be performing every phase.

5. Maintain

Description: The maintenance speaks to the longevity of the system. Is everything operating as intended – a month after being launched, three months, a year, or many years? Important aspects of the maintenance phase include: gathering, analyzing, and monitoring data and feedback, validating design assumptions with users over time, adapting the metrics, and making overall improvements to the system.

Challenges: Continuing to value and place emphasis on maintaining systems can be a challenge. Little was shared by participants on how to do this well in a sustainable way. Who continues to take care of the simulations and update them as needed? What types of analysis and monitoring processes can be performed? How are the metrics improved? For how long should the token engineers be involved in the projects they engineered? Establishing feedback loops with users of the system and simply getting answers to these questions are challenges for token engineers and projects.

Highlights: Economies are dynamic, complex systems. They are alive and something will always need to be changed. Token engineers should also be concerned with providing instructions on how stakeholders can participate in the maintenance of their system.

Excerpts from the interviews:

“(…)It is second, third-order effects of incentive alignment that are kind of unknown until you deploy. And so it's one of those adages of like no plans survive contact with reality. So no matter how well you construct or you understand what the incentives are, there are always people doing things in ways that you don't expect them to behave”. Participant 31

“If you want the metrics to be able to understand how to measure the success or failure of this endeavor, you have to know the individuals who are going to be coming online to participate in this system.” Participant 30

“how do I know how well the economy is doing?” Participant 1

“Once you've created a system, if you find that the system itself hasn't allowed for enough inputs to change the system over time, you may need to re-engineer it. (...) to use that bridge analogy once again, you know, when you drive over a bridge and it was built during the time of the Romans, do you need to knock it down and rebuild it in a modern fashion? [You don't] unless there are constraints that are so changed that you must, in fact, adapt the architecture engineering again.” Participant 30

Process Summary

1. *"What you're looking for is the ways in which individuals can interact with the system and what dangers surfaced. We see some environments where people say, 'Look, I know exactly the kind of individual that will use my product or use this token, therefore, only focus on that behavior'. That's something that in this particular token engineering process that I'm advocating would be less attractive than saying, 'No, but your system allows people to do this. Let's look at all of the ways in which the system will evolve conditional upon the fact that anybody can enter into the system freely and do that.'*

So instead of cherry-picking a behavior that you are sure is going to happen, actually look at the admissible space of actions that can take place and then allow the system to evolve on the basis of anything that can happen. Then say, is the system safe or not? And then of course, it's done normally through a combination of analytical results, if you're lucky enough to be able to write down something that would give you a closed-form solution that's amenable to analysis. More likely it's going to be a simulation where you create a modeling environment that's implemented in a computational framework, and then analyze the outcomes that come from that.

But that analysis then requires the creation of metrics or ways to measure whether the system is in danger or not in danger, and linking those metrics back to what people want, then tells the individuals who have asked for this system whether or not it's doing what they have intended. And so that in essence is a workflow that can be adapted to any token engineering problem, provided each of those stages can be suitably delineated, and I haven't found a system yet where, at least just from the experience that I have, you can't do that. But it does, again, require some of these filters to be passed through where people can understand why you're doing it that way." Participant 30

2. *"We brainstorm what the design might be. And then from there, we go to the validation, which is like, how do I know the design is good, right? What are the metrics that we care about? How do we know it's good? Was there some constraint? Are you sure the proposal that we come up with is implementable or practical? Then from there, we do simulation, and modeling to understand how the different parts fit together. And then from there, we try to build.*

This is where we also work with teams and will actually build the products because it's like at some point being a token engineer be like, 'Oh, we can do all the math and modeling but like, we might as well do it too (in practice). Right?' Otherwise, it's just like in theory, right? So we do all of this stuff as well, prototyping or token engineering. And then from there, okay, you have a live network. So, how do I know how well the economy is doing? So then we also invest a lot in analytics in data.

And then to understand how the network is doing, we work with governance. And then also like, sometimes you look at the data, you also gain more insights about what you can do and you also pay attention to all the advances in cryptography distributed systems. It's constantly thinking about what are some new forms of economy that could emerge that were not possible, let's say, a few months ago or like a year ago, right? So can we do something to really further the entire industry beyond even token economics?

So it's a combination that depends on which card you draw on the day, and then that's the task that we do." Participant 1

Practices

"With the approach you take you set standards that future generations will build up on." - Stella Achenbach

A Day in the Life

"I would definitely prioritize the pathfinding of the individuals themselves, rather than creating one model of a token engineer necessarily, but allowing for token artists to exist. Allowing for various archetypes of designers to exist, and for them to establish their own fields and their own practices and their own templates for others to be inspired by." Cem Dagdelen

When asked to describe a typical day in their practice, **many participants shared that no day looks the same** and that's what they like about the work. They shared different journeys reflecting their varied roles and skills. During the analysis phase of the research, a process called coding was used to capture the essence of what participants were sharing. Action words, or gerunds were identified as codes to help distill down what participants disclosed they do.

These 12 representative examples of participant journeys highlight the action words participants used to describe their daily practices, along with additional themes that were considered worth noting. **These journeys should provide a high-level sense of the token engineering practice** from the perspective of active professionals with varying roles across the field.

- Journey 1 - writing code; writing math; having conversations; running experiments; recording results; educating myself; reading; thinking; foraging every day - Octopus
- Journey 2 - writing and reviewing smart contracts; doing program management project planning; figuring out how to decompose an entirely too fuzzily defined problem into a set of related and solvable sub-problems - Michael Zargham
- Journey 3 - writing; reflecting; balancing solo and social time; attending events; attending educational events; being on Twitter - Wassim Alsindi
- Journey 4 - collaborating; ideating research; articulating; communicating; explaining what we are doing; training; developing knowledge management systems; experimenting with AI; analyzing what happens for people in practice [when they interact with token systems] - Kelsie Naben
- Journey 5 - auditing smart contracts; reviewing a lot of code; questioning if it is the TE's role to review; considering technical and social aspects while reading the text of the

code; reading code; finding mistakes; verifying fault in code or in the design - Jelle Gerbrandy

- Journey 6 - doing operational tasks; communicating; posting on communication channels and social media; finding new projects; serving as a liaison; translating concepts; encouraging more small-scale testing and iteration in tools; imagining the future; seeing tools rapidly evolve; using AI - Jeff Emmett
- Journey 7 - seeing their primary job as a CEO and as coordination reduction; being in meetings; helping connect people; getting out of the way; communicating well - Participant 31
- Journey 8 - identifying as a product owner; coordinating; investing; advising; building tools; enabling the community; wondering what is required in order for someone to be a TE - Griff Green
- Journey 9 - clarifying the various roles that get conflated with the field; researching; analyzing; doing data science; testing; math and coding less than 20% of time; most of time trying to understand the system; hammocking; leading with what you understand from other systems - Danilo Lessa Bernardineli
- Journey 10 - observing and talking with others; immersing in the questions; ideas emerging from conversations; sharing tools and corresponding uses - Andy Tudhope
- Journey 11 - taking a lot of calls; reflecting; researching or familiarizing with client's spaces; learning through conversations; verifying through conversations; sense-making; diagramming; modeling; visualizing; translating - Cem Dagdelen
- Journey 12 - observing; learning; researching; integrating; educating; updating educational materials; partnering; conversing; programming - Angela Kreitenweis

Zooming out and taking pauses

"I use this hammock a lot," Danilo Bernardineli shared, pointing to the hammock visible from the screen of their video chat as an important tool for thinking. **They shared how challenging it is to lack clarity about the object of interest** and that one or two hours in the hammock helps them organize their thoughts by integrating the current problem with their understanding of other systems.

Cognitive functions like thinking, reflecting, and mapping were shared as fundamental aspects for advancing daily work and pausing, zooming out, and taking a systems view. **This goes**

beyond the traditional view of a production-oriented office setting and emphasizes the need to step back and reflect throughout the advancement of the work.

Anchoring on the goal

If there was a North Star that guided people's work, it revolved around clarifying the project's goals and anchoring decisions based on that objective. Participant 28 shared how verification and validation can unveil design flaws between intentions and implementations. **Anchoring on the goal upfront and throughout the process can avoid discrepancies between what was intended and what was implemented, which can also avoid the cost of late adjustments.** Project management skills, including the ability to articulate high-level requirements, play a pivotal role in ensuring that the incentive and mechanism design accurately align with the system's purpose.

Here's what participants shared on the topic:

"Figuring out how to decompose an entirely too fuzzily defined problem into a set of related and solvable sub-problems". Michael Zargham

"A typical day of a token engineer consists of talking to business, talking to legal, and talking to implementation, just to articulate the requirements that are coming from different sides, and to define and translate them into something that can be perceived as a systems engineering lens. So I would say that most of my job is really translation. ...what is the goal, what is the purpose of the system, and then being able to capture those dynamics and capture those requirements into an equation and formalistic language." Participant 23

"A lot of times people come to us and they do not know what they want exactly. They know the pains, and they know their desires, but they do not know how to formalize that or how to fit that into technical tools. So a lot of my time is also trying to do this relationship between the desirable world and the technical world." Danilo Lessa Bernardineli

Summary: Spend more time ~~in hammocks~~, really thinking through the implications of choices.

Pitfalls to Avoid

"In retrospect, now seems obvious, right? Some of these mistakes wouldn't have happened if the systems were tested. Again the Terra collapse, a very avoidable mistake, but we overlooked it and well, paid the cost. Similar thing with exploits like the Mango Protocol. Very avoidable. So I think knowing how to test the systems and establishing standard methods for balancing these economies is an area that I think is very critical, we should focus a lot on that aspect." - Kris Renzo

Despite the infancy of the practice, there are common pitfalls noticed by participants. From the intricacies of economic experimentation in a fast-paced field to the responsibility linked with token design, it becomes clear that defining efficacy in complex environments is no straightforward task. The interview responses underscored the significance of considering broader social implications, a task made all the more difficult by the ever-expanding scope of this domain.

Hasty Scoping and Foundations - The Vulnerabilities that Welcome Hacks

Another critical aspect highlighted is that often, in the rush of starting projects, there's little time to lay solid foundations by thoroughly understanding project objectives, constraints, and parameters. The lack of diligent planning, assessment, and testing was mentioned as a cause of recurring wasted efforts and resources especially when there is pressure to meet deadlines and launch quickly. The consequences of this approach may involve shortcuts that jeopardize the integrity of the system making it vulnerable to exploits and unintended consequences.

A distinction emerged between two fundamental approaches: a **"move fast, break things," trial and error model versus a more structured process**. Jeff Emmett suggests that while trial and error has its place, it should be confined to research environments so that the experimentation is sandboxed and not released for public consumption. This separation in their opinion is considered essential to ensure a clear boundary between learning and experimentation on the one hand and the creation of public-facing, user-centric solutions on the other.

The speaker argues that conflating these two approaches can be problematic, particularly in the context of web3, which often draws from the 'agile' methodologies of software engineering. However, according to Michael Zargham, a shift towards a more careful approach is more akin to civil engineering. This entails developing expertise in a specific set of materials, understanding their optimal contexts for use, and applying them to create reliable solutions for stakeholders without requiring them to delve into intricate technical details.

David Sisson takes a more flexible approach highlighting that different projects may require either high-frequency inputs, akin to the "move fast, break things" approach, or lower-frequency inputs, characterized by a more deliberate design-first mindset. They suggest the key is to acknowledge the potential side effects of choices. In their opinion, the agile methodology approach isn't problematic if the steps given are small and the risks are not big. But when the risks are serious, they advise the use of steady processes, more akin to traditional engineering procedures.

"It's great to break something if you still have your resting on three legs. So, you know, the fourth leg is going out and trying something, and if it breaks, okay, I still have three. But if you're bipedal, and you've picked your leg up, and it's going to land someplace, and that place that it lands is unstable, you know, you can fall over and break something serious. So you have to sort of adapt to the situation that you find yourself [in]. So in situations where the challenges aren't critical, breaking stuff is fine. If somebody's going to die, because something broke, you obviously don't want to go there." David Sisson

Excessive Emphasis on Profitable Projects

Within the realm of web3, it's not uncommon to observe a fixation on projects that promise substantial financial gains. This fervor for profitability can overshadow projects that look for long-term impact instead of short-term monetary returns. Token engineers, while understandably drawn to financially lucrative endeavors, expressed the wish to allocate a portion of their expertise to projects that have substantial social and or technological significance.

"I think a lot of good engineers are attracted to joining profitable projects. So even when we succeed, sometimes we succeed in not the best areas or not the best projects deserving success." Pedro Parrachia

"Lack of technical knowledge is something that pops up out of excessive emphasis on things valued in US dollars. Trying to get rich quickly or trying to be sure that the price of a token doesn't drop, is not something anyone can control." Octopus

Having the right degree of caution and confidence

Danilo Lessa Bernardineli points out two contrasting pitfalls: **the fear practitioners often have of making necessary judgment calls** and **the speed with which some provide answers without proper consideration**. Overindulging in abstract thinking and intricate designs with too much caution can impede progress while diving into solutions without caution can lead to otherwise avoidable mistakes. The challenge lies in finding that delicate equilibrium between caution and confidence.

“Impatience. Not being able to translate ambitious ideas into specific steps. Trying to do too much at once. Lack of focus on one particular area.” Octopus

Clouded Perceptions

Token engineers may find themselves becoming too rigid or stuck in certain tools or methodologies, contributing to a constricted view that inhibits the exploration of alternative approaches to token engineering. **Participant 2 speaks about the need for more resources on first principles and for higher-level concepts to remain open-minded.** The excessive focus on specific tools or methodologies is valuable but can lead to a myopic perspective.

“Being too stuck in certain models of rationality is another problem when it comes to complex systems and systems that are exposed to a lot of emergent behavior, and designing too determinate mechanisms is another, I would say. Tricky.” Cem Dagdelen

*“Draw models on napkins. Napkins are 100 times better than any model is. Particularly complicated software models. I love cadCAD and all of these things, but I feel like you might get a biased perspective, which is why I'm gonna rag on it now. Which is to say that, don't use that stuff, right? Because what you end up doing is like, you put this enormous, emotional, psychological, and intellectual endeavor into learning a new tool, and then that clouds your perception of how valuable the outputs are. Right? Napkins are 100 times better than models **at that early stage** because you can throw them away really easily. And they don't lead to a false sense of hubris about what it is that you're actually doing.” Andy Tudhope*

Understatement of Social Considerations

Many participants highlighted the dangers of operating in isolation from social sciences considerations. Technical excellence alone doesn't guarantee the sustainability of socio-economic systems. In reality, most web3 projects fail because of cultural disagreements among stakeholders. **Although narratives that wishfully picture web3 as fundamental for the betterment of society are so present in the industry, social and relational aspects that could lead to impactful societal achievements tend to be ignored and severely underfunded.**

Kelsie Naben emphasizes the belief, present especially in cryptocurrency communities, that individuals should be able to design, govern, and maintain their infrastructure. They raise a critical question about the considerable time and attention required for self-governance, reflecting on the practicality of this level of personal responsibility. *“Do you really want to govern yourself?”*

This thought poses a crucial consideration to what individuals want to do, and not only what seems to be the right thing to incentivize them to do based on concepts that are theoretically sound.

“What I found in that research or the finding was that the social patterns of these systems are often not considered. Or even if they're considered they're not codified. So people end up with all these bespoke, ad hoc ways of interacting with technical infrastructure without organizing the social aspects of these systems”. Kelsie Nabon

“We think that when we do these kinds of mechanisms, if they don't have any kind of ROI in the next year, or in the next six months, or if they don't, like increase our TVL, over the next one month after they launched something, then they're failures. And I think that's just fundamentally the wrong way to think about it. And it results in really negative personal psychological effects that are difficult to navigate.” Andy Tudhope

Copy-Paste Solutions

Understating the promotion of creativity and innovation within the field is another potential pitfall. The repetition of protocols was one aspect of the theme, but also the fact that teams often don't have scope or don't encourage novel approaches and unique problem-solving, which was considered essential for advancing the field.

“I see a lot of teams that are trying to do the token design, and their process revolves around looking at what other people are doing and copying it without really understanding the implications” Participant 25

“Yeah, I see best people copy-pasting. You just copy-paste and change the name. So there is not much research going on. We will need to spend more time refining our research, and improving the quality of research. We need to improve the quality and depth of our estates. We can't just keep acting on surface-level insights. We need to go deeper into that.” Success D. Ndu

Summary:

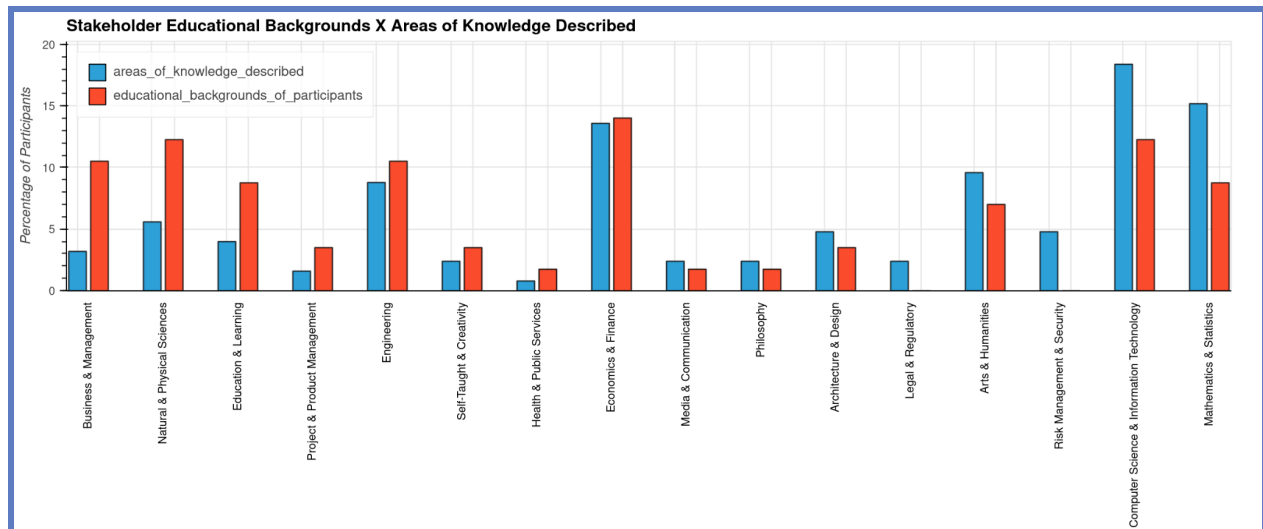
Pitfalls to avoid:

- taking shortcuts that jeopardize the integrity of the system.
- being solely driven by financial motives at the expense of long-term impact and significance.
- becoming too rigid or fixated on specific tools or methodologies.
- neglecting, underfunding, or operating in isolation from social science considerations.

- undermining creativity and innovation.

Areas of Knowledge

Fields that involve complexity tend to be highly interdisciplinary due to their multi-element nature. Interdisciplinarity refers to the combination and integration of knowledge rather than an isolated application of different disciplines. Participants brought up a wide range of topics involved in the practice of token engineering.



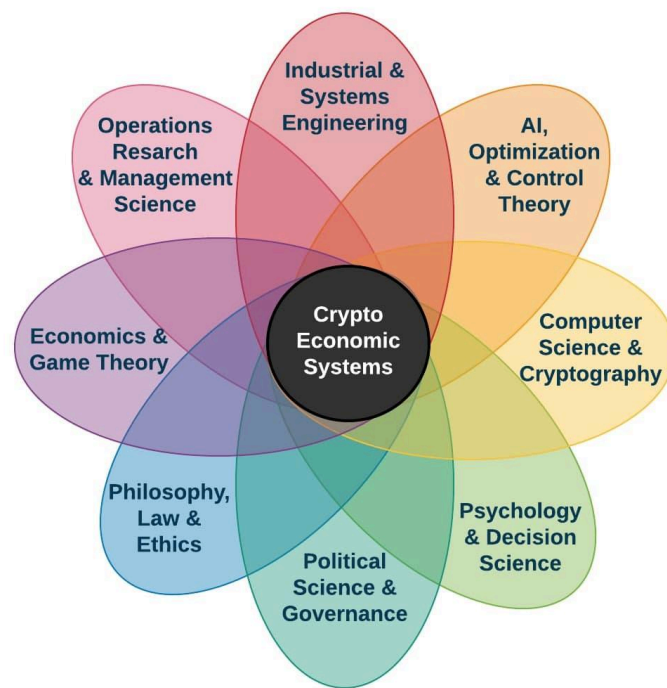
This chart shows the difference between the backgrounds of the participants collected in the consent survey they filled before the interviews and the domain areas mentioned by the participants as being important to the field during the interviews. It's sorted by the difference. Numbers are normalized to percentages. On the far left are the 'over-represented' backgrounds in our dataset. And on the far-right are the 'under-represented' backgrounds.

The core of token engineering lies in the intersection of subjective human perspectives and objective technical possibilities. Mechanism design was predominantly used to refer to the practice of token engineering and it can be described as an approach in economics and game theory that focuses on designing systems with specific goals in strategic settings, considering rational player behavior.

Regarding the emphasis on rational player behavior, participant 12 alerts that rational agents shouldn't be assumed since the psychology of decision science shows that humans are more

than only rational, **emphasizing the importance of a multi-disciplinary interplay, combining social and technical aspects when designing these systems.**

The crypto-economic flower was fairly frequently referred to and is aligned with the areas of knowledge collected in the responses. This list is a compilation of the most relevant areas mentioned and can be loosely grouped into two main categories: exact sciences & technical knowledge and social & behavioral sciences.



Exact Sciences and Technical Knowledge

"Ideally you have a good grounding in math and physics and dynamical systems in particular, you should be able to program in Python and Solidity or some smart contract language and some wrapper language. So smart contract language, the main one is Solidity, but it could be Rust or otherwise. And in terms of a wrapper language, it tends to be Python and JavaScript. You need to be able to do that and you need to be able to work around software systems. If you truly want to be an engineer, you've got to have an ethical bias. So reading the books, and really caring right, having the background there. And then, because you know, these systems tend to be around coordinating humans, it does help a lot to understand aspects of psychology and social science." Trent McConaghy

- **Mathematics:** Described by Octopus as a language of structure and metaphor, providing unambiguous descriptions of structure.
 - **Calculus and Linear Algebra:** Highlighted as important skills for token engineers.
 - **Category Theory:** A branch of mathematics dealing with structures and their relationships. Diagrams are part of category theory and are used in the token engineering process.
- **Computer Science, Data Science, and Physics:**
 - **Computer Science:** The study of computation, information, and automation. It includes software and app development.
 - **Data Science:** The extraction and analysis of insights and knowledge from data.
 - **Physics:** Participants with a background in physics suggested the application of principles from this discipline is useful to understand how local rules and forces can contribute to macro emergent behaviors in a system.
- **AI and Machine Learning:** Likely to play a significant role in token engineering by offering advanced analytical capabilities, automation, optimization, and algorithmic-aided design strategies.
 - Expertise in machine learning and reinforcement learning were deemed beneficial skills to token engineers.
- **Complexity and System Sciences:**
 - **Control Theory:** Studying the control of dynamic systems in engineered processes.
 - **Cybernetics:** interdisciplinary field that studies communication and control in systems, exploring the flow of information and feedback loops to understand and regulate complex processes..
 - **Safety Engineering:** Designing systems with safety in mind.
- **Design and User Experience:**
 - Design, UI, and UX: Falling under both exact and social sciences, concerned with designing interfaces for optimal user interaction.

Social and Behavioral Sciences

"[It is important] to have an understanding of the lineage of how humans went from intention to design to the attempts at deploying those systems in reality and how that worked for society. What their consequences have been, which span the fields of political history, economy, philosophy, anthropology, etc. So all knowledge of all of these fields is extremely relevant. And then you know, when you bring in engineering I think, needless to say, game design, game theory, graph theory, complex systems. Yeah, I would say these are the more tangible practical requirements, and also I guess, anything that's relating to systems and systems engineering"
Cem Dagdelen

- **Economics:** Study of production, distribution, and consumption of goods and services.

- **Finance:** Understanding the management of money, investments, and financial instruments.
- **Mechanism Design:** Crafting systems or protocols that align individual incentives with desired outcomes.
 - **Game Theory:** relevant in incentive design and overall system strategy.
- **Decision-Science**
 - **Social Choice Theory:** falling under both exact and social sciences, is an area focused on building mathematical models of decision-making in social environments to inform governance mechanisms.
 - **Institutional and Policy Design:** The field responsible for the creation of social structures, policies, and agreements.
- **Sociology:** The study of societal patterns and social relationships
 - **Critical Theory:** a philosophical approach that examines societal structures, power relations, and cultural norms.
 - **Qualitative Methods:** Research methods focusing on non-numeric, non-objective data.
 - **Political Science:** Study of political systems, government structures, and political behavior.

*“You need to **understand your audience**. You need to have a certain amount of empathy and acknowledge their needs.” - Stella Achenbach*

- **Psychology:** Scientific study of the mind and individual behavior.
- **Legal and Regulatory:**
 - Legal expertise was deemed a relevant piece of token engineering. Participants highlighted the importance of having Awareness of the regulations from different jurisdictions when dealing with international multi-agent systems.

*“Having an **understanding of the legal landscape** is important because I'm not a lawyer myself, but if I'm making recommendations of economic designs to clients, I need to make sure that I'm not making recommendations that are highly likely to be thrown out as soon as it goes in front of the legal team”. Participant 25*

- **Commons Resources and Elinor Ostrom's Work:** Participants mentioned the importance of understanding the effective management of shared resources without a central authority.
- **Project Management:** Not usually mentioned in token engineering resources but deemed a crucial skill.
- **Web3 Ecosystem:** Participants mentioned the importance of being at par with the latest developments in the fast-evolving web3 ecosystem and its different players.
 - **Market Cycle Awareness:** How the swing between bear and bull markets affects token-based organizations.

*“I think **understanding how the environment can change** that you're building in. I mean, if you're a token engineer, you're building an economy that is going to be in the bear market and the bull market. And if you don't have this perspective of what it is like in those environments, **how can you build a product that will be robust and safe?**”* Griff Green

- **Personal and Interpersonal Skills:**
 - Curiosity and Humility
 - Critical Thinking and Logic
 - **Imagination:** Creativity and innovation for problem-solving
 - **Artistic Expression:** For enhancing imagination and providing new perspectives.
 - **Communication Skills:** Ability to convey ideas clearly and effectively
 - **Real-life Experience:** Participants mentioned the importance of having practical knowledge gained through direct involvement in various situations to bring depth and diverse perspectives to the process.
 - **Adversarial Thinking:** Anticipating and challenging potential pitfalls, vulnerabilities, or opposing viewpoints to strengthen problem-solving approaches and system design
- **Environmental Considerations:** Factors related to the impact on nature when creating new systems and organizations.

Tools

Participants highlighted specific tools that help them perform their work. Most of these tools assist with programming, simulation & modeling, coordination, productivity, and data analytics.

Programming Languages and Libraries

- **Python:** High-level programming language used for simulations, data analysis, and web development.
- **R:** A programming language and environment for statistical computing and graphics.
- **Rust:** A programming language used for systems-level programming with an emphasis on safety, performance, and concurrency.
- **Solidity:** A programming language used for writing smart contracts on the Ethereum blockchain. It provides the means for developers to create decentralized applications (DApps) and implement self-executing contracts with the Ethereum Virtual Machine (EVM).
- **Brownie:** A development and testing framework for Ethereum smart contracts using Python.
- **Hardhat:** Hardhat is written in JavaScript and TypeScript and provides a set of tasks for compiling, testing, and deploying smart contracts.

- **EVMcrispr:** EVMcrispr is a library that can be used to wrap many actions in one vote of an AragonOS DAO.

Simulation and Modeling

- **AnyLogic:** Multimethod simulation modeling tool.
- **cadCAD - Complex Adaptive Dynamics Computer-Aided Design:** A Python library for designing, testing, and validating complex systems.
- **cadCAD GPT:** Open-source LLM agents framework to support token systems simulations.
- **Jack:** Software for human modeling and simulation.
- **Jupyter Notebooks:** An open-source web application that allows the creation and sharing of documents that contain live code, equations, visualizations, and narrative text.
- **Machinations:** A game design tool that aids in creating and balancing game mechanics and systems modeling.
- **Miro, FigJam, and Lucid Charts:** Visual collaboration and diagramming. Also used for the first stages of modeling.
- **Stella Architect:** A software tool for system dynamics modeling.
- **TokenSPICE - EVM Agent-Based Token Simulator:** A mixed signal tool for simulating and modeling token ecosystems using agent-based simulations.

Collaboration and Communication

- **Discord:** A communication platform for communities, groups, and individuals.
- **GitHub:** Code repository and collaboration.
- **Google Hangouts:** Video conferencing tools for virtual meetings.
- **Zoom:** Video conferencing tools for virtual meetings.

Productivity, Project Management, and Document Preparation:

- **Asana:** A project management tool for tracking work and managing team projects.
- **Excel:** A spreadsheet program for data analysis and visualization. Also described as a modeling tool.
- **Google Colab:** An environment that enables writing and executing Python in a web-based notebook.
- **HackMD:** Collaborative markdown editor that allows users to write, share, and collaborate on documents in real time.
- **LaTeX:** A typesetting system for producing scientific and mathematical documents.
- **Notion:** Note-taking and project management.
- **Tableau:** A data visualization tool for transforming raw data into an understandable format.

Data Exploration, Monitoring, and Debugging

- **Dune Analytics:** A platform that provides data analytics and visualization tools for projects on the Ethereum blockchain.
- **TheGraph:** an indexing web3 protocol for organizing blockchain data and making it easily accessible with GraphQL.
- **Tenderly:** A tool that provides insights into the Ethereum Virtual Machine (EVM) and helps debug smart contracts
- **SQL:** A domain-specific language used for managing and querying relational databases

Pay and Incentives

Financial compensation was never shared as the only reason a participant joined or remained in the field. Competitive pay was often listed alongside intrinsic motivation to learn at the edge of a new domain. **The learning opportunities and the belief of being involved at the forefront of socio-economic innovation were predominant sentiments for why people contribute to the field.**

“It’s a fast-moving field that is definitely on the frontier of human knowledge. So the challenge is inspiring. Also, the fact that this can be very long-ranging work. What you do will likely matter in 100 years.” [The main incentives are] the novelty of it, the challenge of it, and then the material compensation possibility.” Octopus

Many participants referenced a sense of purpose and belonging, intellectual excitement and learning, along with their work being in high demand as intrinsic motivators for their ongoing contributions.

Purpose and belonging

There is a feeling of significance to building a different internet. Participants shared they feel immense purpose and enjoy working alongside and feel a sense of belonging with others contributing to the space.

Specifically, participants expressed that creating a positive impact on communities, potentially bringing prosperity and inclusion for those affected, holds deep meaning for them. Many participants shared that their work is helping shape the future and places them in a bigger-picture context that connects them to what they are building and how they are making it happen.

Some participants shared sentiments that their contributions gave them **“the ability to see into the future”**, and **“be part of the ground level,”** and felt like **“moving humanity forward.”** Contributing to this early-stage space excites them, and working with others who share a similar purpose is part of the reward. One participant highlighted, **“You get to work with really smart, interesting people.”**

Intellectual excitement and learning

We are arguably at a point in history where designing economies is more feasible than it ever was, unlike in the past when nation-states predominantly shaped them. Participants reflected on their excitement and acknowledged it as a unique opportunity, sharing: *“If you get the chance to build an economy it's not a small feat.”*, *“Being able to engage at really the bleeding edge”*, and *“you get to redefine money. I can think of very few more powerful things than that. If that doesn't excite you, you probably aren't in the right line of work.”*

Beyond economies, **token engineers are creating social systems**, which demand both a holistic understanding of complex systems and a detailed focus on each of their components. The challenges that arise from the novelty and complexity of the practice are invigorating for those who enjoy being constantly intellectually stimulated.

Participants emphasized both *what* they get to work on and *how* they get to work as ripe for learning opportunities. The transdisciplinary and experimental nature of the field yields diverse perspectives and backgrounds, creating a rich learning environment focused on solving interconnected problems. Skills acquired in this field can also contribute to personal development and insights.

Navigating a high demand for token engineering services

The high demand for token engineers is directly related to the scarcity of professionals with advanced skills. Some participants mentioned facing challenges in finding qualified candidates for senior roles, *“Finding people who will do good work for you is actually in very short supply in my opinion.”* Other participants mentioned the high or increasing demand for token engineering services as a reason to build their skills further. One participant shared that **securing interesting work depends primarily on your knowledge and performance, with many participants having prior experiences in related fields**, which they effectively cross-apply into the token engineering field. There is a high demand for token engineering services, *“we've been offering token engineering services for three and a half years and we've had a continuous stream of clients come to us without any real outreach or marketing”*. In the case of this participant, this was mentioned as something that increased their interest and motivation to continue their work.

In addition to these incentives, participants shared various types of pay and compensation as incentives for their work.

Compensation structures

Compensation structures vary from project to project and can involve a combination of rewards. The participants often accepted the bulk of their payment in a stable token, either fiat or crypto

and have allocations of the project's token or some form of equity as an addition. Compensation types mentioned included:

- Monthly salary
- Hourly rates
- Fixed amount for the entire project
- Equity
- Percentage of the token supply
- Advisory fees
- NFTs/POAPs/ some sort of reputation

It is challenging to pinpoint a precise average pay for token engineering work based on the information provided. It is clear payments can vary significantly, influenced by factors such as experience, geographic location, and the specifics of the project or role within the project.

- Compensation rates mentioned ranged from 25 USD per hour to \$500+ per hour.
- Average salaries for starting junior token engineers were estimated at around USD 70K with senior token engineer salaries estimated at USD 120K+.

Notably, web3 is part of an unstable and fluctuating market, so it is expected that as the industry gains more stability, the jobs related to it will as well.

Participants' perceptions regarding the quality of pay and compensation depended on their context. One participant expressed satisfaction with their token engineering hourly rates, which are 2-3x more compared to how much they make in academia. In contrast, another participant suggested that, compared to similar roles in a traditional Silicon Valley setting, the compensation in token engineering may be lower. Multiple participants advised against entering this space purely for compensation, highlighting that developing skills here can lead to more lucrative opportunities elsewhere.

A participant expressed that the high financial rewards in web3 are associated with the *"move fast and break things"* mentality and not with the more ethical and cautious approach required for effective token engineering. There is also a need to understand the importance of marketing to establish the value of token engineering since, at the moment, financial incentives alone may not be sufficient. While the field is not there yet, there is potential for an impactful *"peer-to-peer network of recognized token engineers"* to be established that would benefit practitioners and contractors.

"I imagine a future where being a member of that quote, unquote, peer-to-peer network of recognized token engineers, comes with its affordances both from your peers and from those who would contract you, but that's a social institution thing. The community has not achieved that status and that's the primary incentive. (...) Today, I think [token engineering] it's scarcely distinguishable from a variety of other buzzwords within the web3 space." Michael Zargham

Project Variations

In the research process, we were curious about the range of projects that token engineers work with. We heard a span from highly technical DeFi projects to more social and communal endeavors. From Bitcoin implementations to blockchain infrastructures to local mutual credit systems.

These differences manifest in the decentralized versus centralized nature of projects, as well as the abstract and conceptual aspects versus the concrete and technical facets. Some projects focus on rigid structure and design, while others remain more flexible. One significant distinction lies in the emphasis on governance and community education, in contrast to more automated projects that prefer to minimize human input. Some projects attempt to balance the profit-oriented mindset and sustainability for the future. The practicality of the token engineer's work can change based on factors including, but not restricted to, the nature of the project they are interacting with, as exemplified by Pedro Parrachia:

"I was involved in a layer 1 blockchain project, a base layer. And I was also involved in a layer 3. The Layer 1 project was very rigorous. We were working with cryptographers, we were working with block masters. We were working with a lot of people to spread out a lot of responsibilities. And I remember how it was fun, but it was also an issue. We had a lot of miscommunication. Then working on a DAPP or Layer 3 is much more flexible, and creative, and requires fewer people. We can iterate faster because it's less infrastructure-focused and more product-focused."

Token-based systems require nuanced design choices. **These key contrast points represent spectrums to consider when working on a project.**

- Technical - social
- On-chain governance - off-chain governance - absence of human governance.
- Abstract - concrete
- Quantitative - qualitative
- Profit-driven - impact-driven
- Long-term sustainability - short-term or finite project duration
- Centralization - decentralization
- Agility - resiliency
- Rigidity - flexibility

Where would you plot your current projects on these spectrums?

Artificial Intelligence

“I know people are really bullish on AI and I am too but I think that the most value will come from really tuning AI systems from the ground up to work with you, rather than just taking something off the shelf and taking a general-purpose model.” - Octopus

While artificial intelligence (AI) was mentioned as “an essential tool”, “a great sounding board”, and “an enabler” that can greatly potentialize the field of token engineering, its role as a double-edged sword is evident, given the responsibility to navigate AI's potential false positives and to ensure its trustworthiness.

AI Helps Facilitate the Token Engineering Process and Navigate Complex Systems

Essentially, AI expands the reach of an individual, enhancing capabilities at various points in the token engineering process. Managing the intricacies and vast amounts of data available in token economies, AI acts as an indispensable partner for each phase of the token engineering process. AI is already being used for **consultation support**, as a **pair programming partner**, along with **assisting in simulation, verification, and documentation work**. With the immense amount of knowledge to take into consideration when building, simulating, and managing a complex token economy, AI seems a better fit for managing this complexity, particularly navigating protocol interactions. Participant 30 shared a potentially beneficial future as, *“having an artificial intelligence that might be able to take a look at the system, the state of the environment and adapt a particular smart contract accordingly”*. For now, nearly all participants agreed that AI is a helpful tool for token engineers, with many already using it in various ways. Most participants also expressed that they did not fear it replacing the work of token engineering, rather it is and will augment their efforts.

AI Accelerates and Expands Access and Learning

There is an immense amount of unknown when it comes to the implications of AI. A number of our participants expressed a desire to move beyond copy and pasting efforts and see AI as solving this by *“recombining information for novel solutions”*. When prompted appropriately, AI can serve as a meaningful thought partner to explore alternatives and challenge biases. A deeper emphasis will be on the verification versus the direct generation of insights, making token engineering more accessible and aiding in the mentorship and onboarding of more early-stage token engineers. One participant articulated the increasing access like this:

“But when we are talking about AI's impact. I think we are thinking about Large Language Models that are recently coming out. I think they of course changed the landscape because now

we can interface with most of these things through language, which opens up everyone even without math capabilities or tooling to start interacting with models.” Rohan Mehta

AI can also be used to review lessons learned from past data ensuring we avoid known pitfalls. Participants believed AI is currently more useful as an assistant, rather than fully trusting the output. Verification of the AI outputs is important yet challenging. Many participants shared an optimistic outlook and believed integrating AI would lead to greater access and resources for those interested in building complex economic systems.

AI, Web3, and Cybernetics

Our participants with deeper AI experience and knowledge reflected on relevant connections for the token engineering field to consider when thinking about AI.

“I think if people had actually studied AI, they'd see a lot of overlap with what we're doing in the web3 automation stuff, because at the end of the day, we have a lot of actuators like points of action in a sense of individuals and smart contracts, lots of sensors in the sense of places data comes from, and we have algorithms that sort of ingest and effectively assimilate the information that comes from all of those data sources into some sort of belief and or action, which is then imposed upon the world and that type of archetype is present in most artificial intelligence systems.” Michael Zargham

“AI is a sort of offshoot of the interdisciplinary research that was cybernetics in the 40s ...Good old fashioned fork.” Michael Zargham

“Cybernetics is a really important word because it's this idea that combines many types of systems, right? ...future systems are going to be combinations of straight-up traditional software technology, plus blockchain token technology, plus AI technology. And there are going to be lots of mixes and matches and overall, they will be truly cybernetic.” Trent McConaghy

Blockchain is the Gift to AI, Not the Other Way Around

The mic drop comes in the not-too-distant future when the relationship between AI and blockchain technology becomes even more evident, as web3 offers an ideal solution for preserving the provenance of AI processes, with all data immutably stored on-chain.

One participant reflected on this interplay between AI and blockchain, highlighting that **blockchain has the potential to bolster AI even more effectively than AI can contribute to blockchain**. *“I think that people are overly optimistic. Yeah, because a big problem in engineering is how you have false positives...in that sense I think that AI is more dangerous than helpful. But if we think of the other way around, which is how token engineering can impact AI. Then I think we can have something really helpful here because one big problem with AI is the question of provenance.”* They continue, *“I think it's going to be a very hot topic. I think that*

token engineering can make a very relevant contribution here because AI models, do not fall under any jurisdiction, like ChatGPT, this question of data provenance is going to be a global issue. And I think that the key advantage of blockchain technology is really that it allows you to have a source of truth that is independent of jurisdictions. ...I think that web3 is the solution for this problem if we want to try to solve it fully.” Danilo Lessa Bernardineli

In other words, there already exists a problem of verifying data origins. Blockchain is poised to help us identify what sources are feeding AI outputs in a transparent way that opens up the possibility of potentially recognizing and rewarding those contributions.

Here are perspectives from the study’s participants sharing how they see AI connecting to the work of token engineering:

*“The difficulty I have with AI tools, especially in the objective domains, is that it is faulty, right? It bullshits a lot. It pretends to know what it's doing. **But your responsibility then as the person using this tool, is to verify if it's correct or not. And sometimes we are unable to understand what we should be verifying.**” Simon de la Rouviere*

*“There is a big intersection between AI and AI-assisted governance for people in the system who make decisions, but also for algorithmic enhancement of automated decision-making within certain parameters. **Algorithms are much smarter than we are but how we design those algorithms is a political question, what the boundaries are and who checks the algorithms and can we check the algorithms (...)** we first need to figure out the basic token design questions that are relevant before we start applying AI to the token algorithms.” Shermin Voshmgir*

*“Some agents can be pure Python. Other agents are a Python wrapper of a smart contract running on EVM. And so you could have it where some of those agents, as you're developing something you can have an agent that's wrapping chatGPT, you can have agents wrapping LLMs, you could have agents that are just simple neural networks or simple support vector machines or otherwise. And you can use AI to optimize things people are doing. **It's more just pure optimization, to optimize, trying to minimize incremental loss** by wiggling weights on balance pools, such that it's minimizing permanent loss for the LPs.” Trent McConaghy*

*“**I don't think it will be like hey, people will lose jobs. Because first of all, clients can't explain what they want.** I mean, the scary part is if people believe it, and start implementing it, but I don't think it'll ever get to a point where you can replace a human in the loop who has experience.” Rohan Mehta*

Summary: AI is already being used in token engineering. It is useful, potentially helpful, potentially harmful, and ultimately equipped to manage super complex systems. Don't count humans out yet - AI is still limited and requires humans to verify, and token engineering might be more important to AI than AI will be to it!

Challenges

“It's always a pain to be an engineer because you want to design systems that are as robust as possible that achieve the best utility for all participants simultaneously. And this is not always possible.” Danilo Lessa Bernardineli

Participants shared the many challenges they face. Regulatory uncertainty emerged as perhaps the most significant hurdle, slowing down the industry's growth and innovation. There are also heavy personal demands, like upskilling and the sheer diversity of expertise required while balancing work cycles and staving off burnout in this demanding field. Communication challenges, both within the industry and in conveying its importance to a broader audience, present additional difficulties. Striking the right balance between simplicity and complexity in token systems is also among the key challenges explored in this section. While there is no straightforward path to resolving these challenges, addressing them is important to the advancement of the field.

1. Standardization

Participants articulated key challenges pertaining to a lack of standardization. Specifically, a lack of:

- terminology standards
- criteria to judge the quality of work
- educational accreditation
- documentation of learnings and case studies
- up-to-date, easily searchable, and accessible knowledge bases

Out of our 41 participants, 25 underscored the crucial role of standardization. Both junior and senior TE's encounter difficulties in accessing high-quality insights such as benchmarks, criteria, guidelines, and frameworks originating from successful and unsuccessful aspects of various projects.

The absence of standards was said to affect the efficiency of the learning curve. Trying to obtain desired information can lead to endless searching, where people end up sifting through an excessive amount of data, including irrelevant and outdated content. Furthermore, the lack of standards was considered a bottleneck for token systems to gain legitimacy in the eyes of regulators and formal organizations.

“The greatest challenge that I have is, everyone uses their own terminology and they make up fancy words” Shermin Voshmgir

"[There is] no established standards, and there is also not yet a huge expert body of knowledge. In the sense that, even from the practitioner's side, it's very difficult to ask an expert opinion or to ask someone what to do in this case, right? There is no literature, there's nothing. There is not even a conference I know of where you could go to and meet people from the space." Participant 23

"Where are people analyzing the things that fail? And the things that are succeeding? Like it just doesn't exist. And it needs to." Griff Green

Although there are resources and events around token engineering, it is clear they are not fulfilling the current needs of practitioners, emphasizing the need for more robust and accessible standards to expand and support the field.

2. Communication

Participants articulated key challenges on the multi-faceted concept of communication. Specifically, challenges arose in:

- translating qualitative narratives to quantitative mathematical functions
- translating general goals into specific metrics
- a lack of a shared understanding about what token engineering is
- clients lacking clarity on what they want or need
- communication between peers throughout the process
- conveying design choices to clients with the necessary nuance
- marketing and creating awareness for what is being built to the broader web3 community

Language considerations in token engineering go beyond the distinction between human-readable and coding languages. Token engineers are involved in various forms of information exchange and are **engineering communication systems**. Participants expressed the challenge to leverage a diverse array of 'languages', including mathematics, data analysis, and visual representations, to establish interaction pathways – each serving as a unique mode of communication and expression. The lack of understanding about the token engineering field, the role of a token engineer, and translating general goals into specific, measurable metrics all further complicate establishing shared understanding and communication. Participants also reflected on the challenge of conveying the nuances of the work and the capabilities of the models and simulations. Effectively communicating was cited as necessary work to produce quality outcomes.

"There are some very specific ways we can help drill down from general to specific, and then the other challenge is, of course, going back the other way. So that once you've got results, you say, 'Hey, look, we want to assess your query based on these metrics, these assessment conditions.' Getting to the understanding of why those are important back up to the top can also be a

challenge because, of course, when people say, for example: 'Our goal is to save humanity', there is this idea that that's an absolute, we want to save humanity, period. And if you say we were able to assess the model and it has a 95% chance of saving humanity, that may or may not print it as 'saving humanity' from the people who are listening. So you have to go through and describe what it is the model is capable of, the simulation is capable of, and the structure that's being promoted is capable of, and that translation is often also very difficult." Participant 30

"It's both so much work to develop these competencies, to do the research and design of those systems and then to do the pedagogy so that people who are downstream of design can understand those objectives and implement them without creating extreme amounts of communication overhead for the organization. It's a huge challenge." Cem Dagdelen

Jeff Emmett refers to the lack of clarity about what is token engineering when they mention that often people refer to "token engineering" or "tokenomics," with a limited understanding, reducing it to a simplistic concept like creating pie charts or deciding how the token allocation is divided. They elaborated that this is a misunderstanding and emphasized that the essential aspect of token engineering goes beyond financial considerations into more intricate aspects of system design. The lack of understanding often leads to assumptions that creating a token system is less costly in terms of time, expertise, and financial resources than it is in actuality.

3. Education and Accessibility

Accessibility and getting people up to speed were among the top challenges to moving the field forward. Specifically, participants highlighted:

- an overwhelming multidisciplinary learning experience
- limited educational resources on token engineering
- lack of educational or career paths for token engineers
- high turnover of tools to use and new releases to track
- limited career opportunities and mentor availability for junior token engineers
- retaining participation and patience throughout the learning journey

Token Engineering resources have evolved considerably since the term's inception in 2018. Tools like cadCAD have contributed to the accessibility of the practice. The Token Engineering Fundamentals course from the Token Engineering Academy has prepared hundreds of students to enter the industry and efforts from the Token Engineering Commons have been democratizing access to funding opportunities for projects at all stages of the token engineering practice. These are just a few examples among many other initiatives that have contributed to the advancement of the field.

However, education and accessibility remain significant challenges. Tools, concepts, and use cases are constantly shifting in this rapidly evolving and multidisciplinary field. As the diversity of

the ecosystem grows, so does the knowledge required for effective work. The pathways available for junior practitioners to attain professionalism are yet to align with the substantial demand for professionals in the field.

Many individuals are eager to acquire advanced skills quickly, driven by the significant demand for work opportunities. Often, they exhibit limited patience for conventional learning methods reliant on incremental growth through work experience and apprenticeships. Despite the noted lack of mentorship within this phase of the industry, Michael Zargham underscores the importance of the learning process, emphasizing that skill development is a gradual process, integrating academic understanding with hands-on experience. Participant 12 spoke of the challenge of helping others understand token engineering concepts and apply them. They pointed out that the main obstacle is the need for individuals to have the right background to quickly grasp and apply engineering principles, knowledge, and tools in the field.

Octopus offers a different but complementary perspective. They advocate for the development of an educational path, where students could become professionals in token engineering without having a prior advanced degree in a related area. *“Despite all of the great educational work that’s being done in TE Academy and TEC and other places, I feel that there’s still a lot of gatekeeping in terms of there’s not a path to coming in with zero background knowledge and then really developing both the knowledge and experience to be good at this. Most of the people I see who are really good at it already have advanced degrees from somewhere. And so they were just kind of translating.”*

Participant 23 sees the transdisciplinarity of knowledge as the main challenge. They argue that this factor contributes to doubt and guessing during the practice, since there are likely parts of the process that aren’t the practitioner’s main knowledge area. These gaps and uncertainties are aggravated by the lack of standards and resources in the field.

4. Funding

The absence of funding models for open-source development incentivizes token engineers to pursue selling consulting services and building closed tools. Angela Kreitenweis highlighted an emerging culture of secrecy around knowledge sources due to the competition and scarcity of resources in the space. Specifically, the related issues are:

- limited bandwidth to develop more sophisticated projects
- stakeholders evaluating mechanisms based primarily on short-term returns
- projects often securing funding to hire a token engineer were considered ponzi schemes
- economic vulnerability to venture capital
- limited scope of one token engineer versus hiring a full team to work on the different elements of token engineering
- lack of funding for infrastructure and tooling

Ensuring that the value created by the project or community is accurately linked to the token, thereby enabling more creative funding mechanisms, presents a significant challenge in the field. The design feasibility adds another layer of complexity, as various pipelines of work are at different stages of development. Some features and tools may be more mature, while aspects like user experience and accessibility are still evolving, contingent on the specific use case. The overall state of web3 directly influences the landscape of token engineering work. Numerous tools are being developed alongside the solutions they aim to support, similar to constructing both a bridge and the machinery needed for its implementation simultaneously. In addition to securing appropriate funding for the token engineering process in projects with clients, participants expressed the difficulty of securing resources for developing infrastructure in such a convoluted stage of the field.

“They want to hire a token engineer to design their whole system, without recognizing that that’s probably a whole industry with a pipeline of different firms.” Jeff Emmett

“There is a lot of behind-the-scenes shit that people do to make a token economy work, like market makers. That’s where they pay market makers to trade the token on exchanges and get enough investments so that there’s going to be liquidity just because the VCs know they need to maintain liquidity until they can dump.” Griff Green

“I think a lot of them had really bad token models, basically Ponzi schemes, in some sense I mean, maybe that’s a bad word, but it is practically true. They were just trying to speculate quickly, like shooting their token up and thinking about quitting and you could hear that in the way they were talking about it. So, I felt like that was a personal challenge to figure out which ones I should work with and which ones I shouldn’t, especially when I was not getting good options...Ethically I was very confused as to what I should be doing in any of those scenarios”. Rohan Mehta

5. Complexity Meets Infancy

Some of the challenges that emerged from both the complexity and nascency of the field were:

- innovation happens on multiple layers simultaneously making it hard to track inputs and outputs
- with so many variables it is difficult to identify why something is and is not working
- buggy user experience with inherent risks - the systems are not for everyone
- short attention spans and lack of follow-through
- Even with models and simulations, there are many unknowns about how people will behave once the system is active

Token engineering is still in its early stages, with immense untapped potential, especially in the realm of incentive design. The field is progressing towards academic formalization, while the capabilities of web3 continue to expand exponentially. Navigating the challenges posed by any

nascent field can be daunting. The multidisciplinary nature of token engineering, coupled with its focus on socio-technical complex systems, adds layers of depth and intricacy to the challenges posed by its early stages.

The work of token engineering deals with complex systems, which are often characterized by a collection of interconnected and interdependent components that, as a whole, exhibit behaviors and properties not directly predictable from the behaviors of the individual parts. In a complex system, interactions among its components give rise to non-linear and emergent phenomena, which are outcomes that are not explicitly programmed or designed into the system. This highlights the complexity and unpredictability of the work.

*“It always happens with new technologies and new technical paradigms that until we have the experience, the systems can be a little bit erratic, or unreliable, and behave in unexpected ways. (...) So I guess the biggest challenge is **we understand neither the socio nor the technical part of the socio-technical system that well and only with time, only with experience**, we will actually make those progressive gains and advances that we can build this into a more mature and a more, let's say, self-confident field.”* Participant 4

“it's easy to tell, I guess when something is working, but it's hard to know for sure why something's working” Participant 2

Jessica Zartler adds that the fast pace and multidisciplinary nature of the field often lead to quick change of interest, difficulty in finishing projects or analyzing their longer-term impact, and a general sense of burnout from this intellectually intense environment.

“We are (now for token engineering) like 20-25 years ago (where developers were) for programming websites” Angela Kreitenweis

6. Regulatory Compliance

When sharing what keeps them up at night, the regulatory landscape was the most pressing concern participants expressed. Mostly due to:

- unclear risks and consequences for token experiments
- liability concerns, specifically on who can become liable for multi-stakeholder systems
- regulatory constraints resulting in a slowdown of the industry
- too high of risk as not all projects can afford the risks associated with token innovation while ensuring regulatory compliance
- insecurity in the design phase leading to unanticipated or delayed costs
- limited access to domain expertise in law for blockchain-based experiments

Navigating the intricate landscape of regulatory compliance presents a complex challenge for token engineers, influencing critical decisions at the intersection of innovation and legal frameworks.

This delicate balance often translates into practical challenges, where well-conceived token engineering projects may encounter delays and heightened risks due to regulatory ambiguity. Numerous instances were referenced where rigorously developed models encountered obstacles due to legal uncertainties, resulting in roadblocks. Consequently, the industry's pace and trajectory have experienced a notable slowdown, as expressed by Matan Field:

"I think good token engineering design could impact so dramatically, the world so widely. And, and I think it's slowed down, I don't know, maybe 10x, just because of regulation."

An **"anti-token sentiment by regulators"**, as voiced by Participant 22, might further complicate matters, adding a layer of apprehension even to meticulously designed projects. The struggle to harmonize innovation with regulatory constraints underscores the pressing need for clearer guidelines and collaborative efforts between legal and engineering realms to ensure a smoother journey for token engineering endeavors.

"You know, we can build very beautiful models, fantastic models, but the lawyers say that it's not possible because it potentially might one day somehow, in the future, become a security or it might be, it might be a problem with the regulation" Participant 22

Summary: Regulatory uncertainty and communication barriers hinder the field's development. The absence of standardized practices and limited educational resources exacerbate the overhead, while funding models pose additional hurdles to innovation and industry growth. Despite the potential, token engineering remains in its early stages, with no shortage of challenges to address.

Needs

Participants shared what is required for the advancement of token engineering and web3. The Needs section is informed by the insights gained from the Challenges section. It emphasizes the importance of establishing libraries of knowledge and standards as foundational elements, which serve as the necessary groundwork for education, particularly mentorship, and the development of a mature social infrastructure.

1. Libraries and Sandboxes

The interview data depicted a need for a library of knowledge. Specifically, a collection or repository of protocols, models, primitives, and mechanisms.

- *protocols* refer to guidelines for information exchange between entities,
- *models* would encompass conceptual, mathematical, and computational models,
- *primitives* are the building blocks that form the basis of system functionality, and
- *mechanisms* refer to in-depth explanations of underlying components or processes that execute actions specified by protocols, models, or primitives.

While significant progress has been made in the creation of token systems, research is absent regarding the methodologies, successful strategies, failures, and lessons learned from numerous past and ongoing experiments. So, **the needs are not only related to the compilation of these resources but also to detailed analysis of how they were used and insights about their implementation, achievements, and safety over time.**

“There are people that know what they're doing, but what they're doing is changing all the time”
Participant 15

To create an effective library of knowledge it is necessary to **collect and store quality data** that is both a) inclusive of data apart from what is on-chain, like stakeholder and market behavior, and b) dynamic and reflective of the evolving practices within the field. The need for a library of knowledge was underscored by a few participants sharing frustration at the lack of shared information and resources, like Ataberk Casur who shared, “there needs to be more research outputs. I was just researching staking contracts yesterday. And I couldn't come up with an article where it just listed all the various ways staking is used by protocols as an engagement method.”

Having repositories of technical and conceptual resources paves the way for sandboxes where hands-on exploration and experimentation with various models, primitives, and mechanisms can happen in a low-cost environment. This benefits the field in many ways, minimizing the consequences of bad design choices, which can be very expensive and harmful to stakeholders. Therefore the goal of building libraries of knowledge is to also heed the call of

many of the participants, which is to develop sandboxes. As Participant 35 put it, we need places *“where we can experiment and fail safely, in a way that doesn’t put too much capital at risk, or cause reputational damage”*.

Regarding experimentation, Rohan Mehta expressed the need for *“some kind of plug and play system through which you can just look at some already made token systems that people have applied or AirDrop systems that people have applied, be able to like just pick and drop and say that okay, if I do this, what will happen if I do this, what will happen?”* Other participants shared similar desires to have **libraries that could be interactive, offering some sort of simulation environment for testing the modules they are interested in**. Better libraries mean improved modularity for experimentation, which improves design outcomes.

2. Standards

A library of knowledge would perhaps be the first building block of a broader need which is the establishment of standards. Standards refer to criteria, specifications, or guidelines to ensure consistency, interoperability, quality, and safety in various token systems and their engineering practices.

Simon de la Rouviere advocated for **provenance** and the contributions it can bring to standardization. They questioned the existence of documentation about projects considered successful and the specific metrics or criteria used to evaluate and define success within the given context, *“what have been the sort of success stories that we can say, like these things succeeded, or at least sustains itself? And what are those success metrics? How are we defining that success?”*

Standards act as a benchmark for assessing and categorizing systems. This ability to judge the quality of different systems based on shared guidelines throughout the industry can support the creation of

- a. **Academic Programs:** Academic programs support the long-term sustainability of disciplines, providing ongoing education research and development. Interviewees expressed the desire for an academic community including peer-reviewed journals. Participant 23 shared that without the establishment of standards, the field will never mature. They defend the creation of academic programs and state that for development *“You need to have people collaborating on this together, test and validate both standards and methodologies. And this is what we need, but this is a long-term goal and long-term project.”*
- b. **Regulatory Compliance Advancement:** Regulatory compliance was specifically shared as a topic of concern keeping token engineers up at night. Participants suggest expanding lobbying groups and knowledge sharing about different jurisdictions and their limitations, including more targeted education, as Matan Field suggested, *“We need to*

have the industry educating regulators about this, and somehow finding jurisdiction that can experiment with that in a healthy manner.”

- c. **Token Engineering Audits:** Briefly covered in the process section, these audits could involve a set of criteria validated by various stakeholders in the field. This framework can apply positive pressure on projects, encouraging them to perform an audit of their token systems before exposing them to the public. Griff Green voiced concern about the absence of a system audit in a hypothetical future where these measures were expected, stating: *“What, you didn’t do a token engineering audit? You didn’t have anyone review this system?”*

*“We all know we’ve driven over a bridge even if we don’t know how it works. We trust it. We trust the people we never will meet never have met in high probability of likelihood. And yet we drive over these bridges and we say this is good because there’s a whole standard that’s been set up and there’s of course oversight that allows us to ensure - knock on wood - that these bridges maintain their stability. Of course, we all know tragic exceptions that do occur, but they serve only to reinforce **my feeling that at least in the token engineering environment, we really are in the step now, where if we fail to place standardization at the top of one of our priorities in the near future, we will find ourselves in an ever more, let’s say unfortunate series of circumstances** where we see projects that lose a lot of value and a lot of safety for individuals.”* Participant 30

Overall, standards were mentioned as an important factor in improving education, interoperability between projects, communication between peers, the quality and safety of designs, and the legitimacy of token engineering as a discipline.

3. Education and Mentorship

More than any other suggested need, 75% of our participants highlighted the importance of advancing token engineering education. Creating and expanding formal academic programs, constructing certification criteria, and offering accessible educational resources and support were cited as primary needs. Each of these needs has been elaborated on within the desire for a library of knowledge and standardization. However, there were unique insights shared about how to support the advancement of token engineers spanning from beginner to advanced levels through mentorship and educational support.

Mentoring was mentioned several times as a fundamental practice to help junior token engineers level up. Octopus, brought the perspective of a professor and the lessons from traditional institutions' mentorship practices, *“one of the big things that a traditional institution would offer is the fact that at some point you get to apprentice directly with someone who has achieved what you want to do. So opportunities for apprenticeship and shadowing would be really good and also making sure whatever the objectives are, they clearly focused and communicated.”*

While the importance of mentorship is well understood, the **scarcity of experienced mentors** emerged as a bottleneck, in addition to a lack of incentives for senior token engineers to spend time mentoring as it requires more time and effort in many cases. Michael Zargham, put it this way, *“People who are aspiring token engineers, need to continue development of educational material at more advanced levels. They need access to projects where they can serve as junior engineers and develop skills with people who are willing and able to teach them and to have the time to do that because there are some incentive issues there. It does make projects sometimes longer and harder to support junior engineers. It doesn't always make it easier.”*

Participants also expressed a desire to have more native pathways to upskill in the field and reiterated the precursory need for standardization and a library of knowledge to facilitate that outcome. *The Vienna University of Economics and Business* has maintained a *Cryptoeconomics Research Institute* since 2018, actively organizing events, and seminars, and supporting the production of academic papers on the theme. Several participants also cited the *Token Engineering Academy*, which has onboarded hundreds of practitioners to the industry, as an example of an open and native pathway for token engineering education. Although there is still a considerable journey ahead to meet the ecosystem's educational requirements, more mentorship and access to opportunities sit atop the list of educational needs alongside the foundational work of libraries and standards.

4. Social Infrastructure

Technological innovation often overshadows the social elements ingrained in any system that involves humans. Several projects have failed due to a lack of attention to the subjectivity of relationships and culture. Token-based systems demand extra consideration, especially when projects are implemented on-chain, as rectifying issues later in the process requires a significantly greater amount of effort.

Success D. Ndu shared their perspective on the importance of forging social connections not merely for networking's sake but as a strategic step in steering the industry beyond short-term gains, *“As a community, we need to start driving hedge funds, and intelligence firms, control protocols, to enrich token engineering and embrace simulations as scenario analytics, to be able to predict how users interact with their platform, and predict how the system scales over time. Is this system scalable? What do we need to adjust? What are the guardrails we need to install into this product to make it sustainable over time? We can't as an industry just keep churning out ponzi's upon ponzi's upon ponzi's. Nobody's going to take us seriously.”*

Participants shared concern about the insular and narrow nature of projects in the space, often citing it as a human failure to establish meaningful relationships and discussions about the problems-to-be-solved within society. Pedro Parrachia outlined the work ahead, *“We need to make an effort to burst the bubble to reach out to more legacy organizations, to joint projects and operations, not only with legacy organizations but also DAO to DAO. There's so much DAO*

to DAO stuff to be done. We're progressing slowly there. So I think [the need is] paradoxically coordination."

In this regard, better human coordination is a worthwhile focus and is far from being an unsolvable problem. According to participants, the following are important social infrastructure aspects needed to advance the field:

- a. **Partnerships:** Fostering an environment of cross-pollination, where partnerships can flourish, might seem time-consuming in a fast-paced industry where practitioners are in constant high demand. The horizontal tendency of web3 can also bring a sense of scatteredness, where it's not clear who is championing what and what work is being repeated due to a lack of communication. We have heard in the interviews that creating opportunities for knowledge sharing is laborious but rewarding. It pays off in the long term by preventing double work, opening insights for problem-solving, and strengthening the industry as a more robust body. Participant 22 identified a gap in roles focused on this essential coordination, *"There needs to be a lot of partnerships with other people in the space. And I don't know who is the right person championing that?"*
- b. **Productive Criticism:** Another theme from the interviews is the need to establish a psychologically safe **environment where stakeholders can challenge the token engineering process**, including feedback and making space for failures - even highlighting failures so everyone can learn from them. Participants highlighted that uncertainties and lack of knowledge, standards, and principles are currently prevalent in this space. More open dialogue and disagreement, where criticism is focused on the process and ideas, not on the people, is desired to enable a more robust design process both for experts and for new talent. However, Participant 4 cautioned that this does not mean having tons of meetings for people to weigh in, *"And it's about coordinating reduction. Because if you have multidisciplinary people, they can balance this out, rather than having 15, 20, 30 meetings, being able to like, teach and acknowledge, you know, so I think that more proper kind of training."*
- c. **Conflict Resolution:** There are numerous cases of conflicts being a decisive reason for project failures. Elinor Ostrom was an economics Nobel-winning scholar and was mentioned in a portion of interviews for her work on institutional design, governance, and commons management, which includes a preoccupation with conflict resolution. David Sisson shared this is an area that needs more focus and effort by stating, *"The mechanisms that are being engineered, need to be able to maintain organizational coherence in the face of essentially conflict. You know, it's not all just cooperation."*

Additional perspectives were provided regarding the use of social infrastructure to include more social sciences in the end-to-end process and to give projects an impact orientation, both aimed at enhancing the human experience and outcomes.

Blockchain technologies open even more possibilities to improve human collaboration and coordination. Building social infrastructure highlights the human coordination necessary to advance cultural and technical practices, enlarge possible use cases of token systems, and create more responsible and sustainable working environments.

Summary: Foundationally, we need to develop libraries of knowledge, sandboxes for experimentation, and standards. The acceleration of effective learning environments requires a focus on more mentorship and social infrastructure to promote productive criticism and relevant relationships to ensure quality outcomes.

Ethical Considerations

“Every new technology brings in new moral questions. (...) I'm an engineer. I am tasked to increase engagement for my platform. The second-order effect is that it affects people's mental health. I as an engineer did not see that coming - where the algorithms that I was contributing to elicit hate or aggression. And I think that we can learn from that in terms of token engineering. This is why we need to be able to take a step back and this is why ethics is hugely important because it's going to bring up new moral questions of what direction we go and it can affect a lot of people.” Participant 31

Participants expressed varied perspectives regarding the inherent subjectivity in determining the ethical morality of actions within token engineering. Especially when designing socio-economic systems that involve managing financial resources, terms like "right" and "wrong" were regarded by a few participants as too subjective to use. What one person may deem wrong, another could see as right.

In this section, ethics refers to a systematic and communal reflection of principles, rather than a space to emphasize personal sets of beliefs about what is right or wrong. David Sisson clarifies that ethics “is that thing that allows you to discuss the rightness and wrongness of something.” The intention is to open an analytical exploration of the ethical deliberations and considerations that emerged through the course of the interviews and shed light on the variation in perspective, emphasizing **the crucial role of ethics in token engineering despite the challenges in achieving consensus or definitive ethical guidelines.**

Perceptions on Ponzi schemes

Participants shared experiences of being approached to develop ponzi designs, illustrating the ethical dilemmas token engineers might encounter. Discussions on this theme revealed a spectrum of opinions: some underscored the inherent moral conflicts associated with ponzi designs, while others contended that such schemes could be ethically defensible, provided there's full transparency. The next paragraphs elaborate on different elements that emerged regarding the ways different stakeholders navigate such dilemmas and the nuances of ethics in TE.

“Ponzi-based designs are an area where there can be a potential ethical mismatch between what a particular company may want and be asking from a token engineer to create incentives that maximize revenue within the system. And that may come from creating incentives that are incentives for activity greater than what an agent would naturally spend, based on the fundamental product itself.” Participant 25

“You also catch a lot of users that don’t fully understand the risks of ponzi-structured products and can get badly burned by them. And that is a big source of ethical risk for those who are designing systems that have the potential to be ponzi structures.” Participant 25

“I think it's ethical to do a transparent Ponzi.” Griff Green

Responsibility and safety in complex environments

*“For a token, is this safe? I believe **token economies are public infrastructure**. It's the same as a bridge. It's just a financial bridge. It is a public infrastructure that anyone can cross that bridge; anyone can buy Luna, and anyone can buy into an ICO. And usually, the token engineering is done in a way where we need to protect the people who are launching this project. But **who's protecting the people who will use the project? And I think token engineers are the only people who are there to do that.**” Jessica Zartler*

The nascent state of the industry, combined with its constant evolution, presents challenges for practitioners. These challenges include navigating the nuanced spectrum of risk, which affects not only the practitioners themselves but also their clients and users. Additionally, there is difficulty in understanding the levels of knowledge available, evaluating both what a user knows and what they are capable of knowing.

A key liability for the Token Engineer is related to the information asymmetry between users and engineers. Users are not usually expected to comprehend the entire system, while the engineer is. This creates a responsibility for the Token Engineer to ensure that important perception and actuation gaps are accounted for. However, there is also a challenge in determining what constitutes a safe and healthy environment for both users and engineers in such a nascent field.

The lack of standards and certification criteria in token engineering contributes to confusion over the responsibilities of Token Engineers and the areas of action for which they are liable.

The potential **risks of engaging with token systems can range from financial loss and reputational damage to regulatory complications and numerous unknown variables** in between. While acknowledging the responsibility of token engineers in ensuring the safety of stakeholders interacting with the systems they design, as emphasized by Griff Green, *“This is why we’re here – to protect the public from potential issues,”* some TE’s expressed uncertainty regarding the level of responsibility attached to their roles. Participant 30 uses the analogy between token engineering and the engineering of bridges to clarify the responsibility of engineers when it comes to ethics, *“You would want the bridge to be safe structurally, that’s step zero, but you also want to ensure that you’re not doing something in the bridge that is ethically wrong. Like for example, keeping out certain individuals or certain types of people or ensuring that there is a collapse of diversity or inclusiveness.”*

It becomes apparent that the term "engineering" highlights the emphasis on safety but is not enough to ensure an ethical stance in the design of systems. When asked about diversity and inclusivity, Jessica Zartler shared a story that serves as an example of protecting the public. They recounted an incident from the 1960s in New York City when a bridge was being constructed in a high-end area. Project owners lobbied engineers to build the bridge too low for buses to pass because, in that context, buses represented less privileged classes. The height of the bridge was deliberately limited to exclude that group from accessing those specific beaches in New York. This was a form of discrimination, as the beaches represented a public good made exclusive by the engineering process. This example, along with others shared by participants, illustrates that ethics isn't only concerned with the safety of public structures but also with their accessibility.

Summary: Overall, participants expressed concerns about the potential for abuse and exploitation of the systems and urged for mechanisms to prevent the creation of systems susceptible to misuse. Addressing biases in the design phase, and avoiding blind replication of mechanisms were considered essential components of promoting safety throughout the token engineering process.

Is Transparency Enough?

In the realm of web3 systems, transparency can be understood in a variety of ways and intricately linked to ethics, incentive structures, and privacy concerns—a tension frequently echoed by our participants. There is an emphasis on viewing transparency not merely as the availability of information but as its **accessibility and comprehensibility**, highlighting the ethical imperative for information to be not only open but also understandable to all stakeholders. Additionally to the accessibility concern, participants named transparency and privacy as central tensions in web3 that are being addressed by cryptographic-based technologies like Zero Knowledge Proofs. Additional efforts are needed to continue building transparent, accessible, resilient, privacy-preserving, and agile systems.

Participant 2 shares that **designing incentives can be unethical depending on what stakeholders are being incentivized to do**, but they stress that as long as the incentive structure is transparent, then stakeholders can have autonomy. They add *"You shouldn't say it's decentralized, if it's not decentralized, or you shouldn't say the token supply is capped, if you have the admin keys and commit more tokens and stuff like that. But then as long as you are transparent, people can choose whether or not they want to participate in your economy"*.

Considerations of the audience are important to designate the level of accessibility information should have. On-chain data, Github repositories, forums, and extensive documentation might be available abundantly but if they are not easily understood by the people who are interacting with those systems and products, it can become a problem. So what could be a satisfying direction for having transparent and digestible information that prevents manipulation?

Participant 2 exemplifies the harmful impact that the lack of accessibility can have: *“It’s possible to design systems in a way that works for a short period of time and can get people excited but then, ultimately come crashing down. And that’s probably one of the more dangerous things and when you’re dealing with stuff that’s super complex, like the Terra Luna thing, so many people, and they were in many ways transparent about how it worked, but it was very complex and hard for the average person to understand exactly. And so because of that, a lot of people kind of just trusted it or saw that it was working and had a large amount invested in it. So I assumed it was all fine and dandy. But sure enough, like the way it was designed was not actually sustainable and had some critical flaws and probably unethical to market it in a way that made it feel a lot safer than it actually was”.*

Building on this quote, the importance of effective and accessible transparency becomes evident. It would open spaces for discussions, offering a collective pathway to navigate complexity towards safety, as opposed to the isolated and blind trust that exists with inaccessible transparent information.

Conversely, privacy considerations, avoiding system exploitation, and protecting people creating and using the systems seemed to be in contention with the available, accessible, and easily understood focus as highlighted by Participant 22 *“When it comes to engineering, token engineering, especially, you can financially engineer a token structure to extract a lot of value. And that is scary because when we’re doing token engineering, a lot of things are public, publicly available, people can see how things are being structured, and people can exploit the system.”*

The discussion herein underscores a complex paradox: the necessity of transparency for ensuring ethical token engineering and stakeholder autonomy clashes with the imperative to safeguard privacy and prevent exploitation. This dilemma involves dealing with identity systems and calls for a nuanced approach to information sharing—one that ensures clarity and accessibility while also guarding against the vulnerabilities that open information may expose.

Summary: Transparency alone won’t uphold ethics. Accessibility is essential to promote inclusivity and safety.

Beyond Code: Context and Consequences in Token Engineering

An essential aspect emphasized by interviewees was raising awareness about the broader implications of token systems. The broader context in which a system is implemented can inform the token engineering process to avoid disruptive cultural or political consequences. Participants also questioned the ethics of the *“move fast break things”* design approach, considering the financial and emotional risks users get exposed to when interacting with these systems. They also stressed the need for adaptable structures and the placement of ethical guardrails to prevent unintended consequences.

“It's important to always consider the broader impacts that a technology can have in terms of its externalities, or even just disruption in itself, like, disruption can be positive, but it's still disruption in established norms, practices and relationships.” Simon de la Rouviere

Beyond their technical applications in consensus algorithms and automated processes, tokens hold a transformative potential as coordination tools for addressing significant social and environmental challenges. This potential extends the ethical considerations of token engineering into the realm of global impact and responsibility.

Angela Kreitenweis illuminates this broader perspective by citing the urgent issue of climate change. **They advocate for the utilization of tokens as instrumental tools in navigating the complexities of this global crisis, showcasing their capacity to venture into uncharted territories that traditional currencies cannot reach.** This perspective is grounded in the belief that tokens can play a pivotal role in mobilizing resources and coordinating efforts to adapt to and nurture local potentials, rather than exploiting them. Here, 'local' refers to the unique social, economic, environmental, or cultural dimensions of specific communities, underscoring the need for tokens to be designed with a deep understanding of and sensitivity to these diverse contexts.

Aligning with this view, Kris Renzo stresses the importance of ethical considerations in the development and deployment of token systems, underlining, *"We want to have people who care about what they're building and who they are building it for."* This statement highlights an ethical proposition encouraging token engineers to prioritize the well-being and interests of the communities they aim to serve, emphasizing the positive impact of token systems.

Michael Zargham voiced concerns about the investment of time and effort required to consider stakeholders' interests, understand the broader context of different projects, and devise appropriate solutions that are dedicated to mitigating potential harm. The challenge emphasized by them lies in allocating sufficient budgets to support these comprehensive considerations. This level of dedication to stakeholder care and ethical solution development often materializes only in scenarios where those financing the projects (the demand side or payer side) recognize the importance of these factors and are prepared to cover the costs associated with upholding such a standard of care. This situation underscores the critical role of financial support in enabling token engineers to perform their duties responsibly and ethically, ensuring that solutions are thoughtfully tailored to the unique needs and challenges of each context.

Summary: Context matters. It takes time and intentional efforts to consider the broader impacts that a technology can have in terms of its externalities.

Applying Guiding Principles

“What's definitely missing in the field is some kind of analogous to the Hippocratic oath” Danilo Lessa Bernardineli

This comparison underscores the absence of universally acknowledged ethical standards within token engineering, highlighting the field's nascent state and the necessity for a more structured approach to ethical considerations. This absence is not merely an academic oversight but a practical challenge to ethical navigation within the field. The diversity of views on what constitutes an ethical practice when developing and implementing token systems points to **the complexity of defining and implementing ethics in a domain that is as varied in its applications as it is in its stakeholders**. The call for multi-stakeholder conversations and collaborative research efforts, as deemed important by participants and elaborated in this report in the 'Needs' section is also relevant in this context. These collaborative endeavors aim to forge consistency in ethical standards, ensuring they are designed with consideration for the broader impact on society and the environment.

The analogy drawn by David Sisson vividly captures the contrast between traditional engineering disciplines and the relatively uncharted territory of software and, by extension, token engineering: *“NASA doesn't want rockets with people in them that blow up. They apply engineering principles. You don't want a nuclear reactor to go critical on you. So you apply engineering principles to that. Even a simple industrial site needs to be properly engineered to make it as safe as reasonable. **But my worry with software engineering is that everything sort of gets the treehouse treatment** where you know, ‘Okay, I got a great idea, let's go build a tree house.’ And so you just do it. Sometimes the tree house stays up, sometimes it falls down. So hopefully you haven't built it up so high that you get hurt when falling down. That's the difference.”*

The meticulous application of engineering principles in fields such as aerospace or nuclear energy, where the stakes of failure are well known, contrasts with the often experimental and rapid-iteration approach prevalent in software development. A "treehouse" mentality mentioned by David Sisson —where projects are pursued with enthusiasm but perhaps without sufficient foresight or accountability for potential consequences—highlights the need for guiding principles that can instill a culture of responsibility and safety akin to more established engineering practices. Establishing such guiding principles is envisioned as a step toward maturing the field, where ethical considerations are integrated at every stage of the engineering process, mirroring the rigor and responsibility observed in other engineering disciplines.

Summary: The discussion around guiding principles in token engineering is not merely about the formulation of ethical codes but about fostering a perspective shift in how projects are conceptualized, designed, and implemented. It calls for a reevaluation of the responsibilities of token engineers, urging a transition from a focus on feasibility and innovation to an equally strong commitment to ethical integrity, stakeholder welfare, and societal impact.

Sustaining Contradictions: Spaces for Dialogue & Disagreement

“What is ethics? It’s a very interesting question to unfold in dialogue with other people who have different priors, and conditioning, and come from different backgrounds, cultures, beliefs, races, sexualities, religions, and all of this. In the modern world, there is no universal ethic, I think there are really interesting things that we can point to which seem to go across most cultures” Andy Tudhope

The exploration of ethics within token engineering reveals a landscape rich with contradictions and the need for sustained dialogue and disagreement. Andy Tudhope’s reflection on the nature of ethics emphasizes its complexity, noting that one of the ways in which ethical practices and principles can emerge is in dialogue among individuals with diverse experiences, beliefs, and cultural backgrounds. Such an understanding of ethics highlights the importance of creating **spaces where differing viewpoints can be expressed, explored, and respected.**

David Sisson speaks of the capacity AI has to amplify complex system engineering analysis allowing more people to participate in analytical functions, however, this also multiplies the risk involved in those activities. They state: *“To err is human to forgive is divine, but to really screw something up takes a computer, you know, we’re still in that state. So we can really screw it up now. Because computers are really powerful. In that sense, I think ethics plays a huge role. Or at least it needs to.”* The power of technology to amplify the consequences of our actions—both good and bad—is poignant. David Sisson reminds us that in the age of machine learning, the stakes for ethical decision-making are higher than ever. This amplification makes the role of ethics not just important but essential, as the potential to *“really screw something up”* with technology demands a greater responsibility for ethical foresight and consideration.

The interviews collectively expose the **critical nature of long-term thinking** in ethical decision-making within token engineering. There is a concerted emphasis on learning from past mistakes to foster responsible technological advancement. The dialogue around ethics in token engineering goes beyond navigating present challenges towards laying a foundation for future innovation that is conscientious and responsible. As reflected by Success D. Ndu, this involves balancing the interests of diverse stakeholders while prioritizing the long-term success of the protocol, which will likely require navigating trade-offs. Griff Green’s comments further reinforce this sentiment, illustrating a commitment to developing systems that are safe, durable, and designed to address real-world problems:

Summary: Embracing contradictions and promoting dialogue among diverse stakeholders is fundamental to ensure the sustainability and relevance of token systems.

“And I think especially the people really designing most of these systems, at least in the Ethereum space, most of them actually care about that. They want things to be safe and they want them to work for a long time. They’re building something. They’re not building a scam

project that's just trying to make money. They're building something that's solving a real problem." Participant 3

Diversity, Equity, and Inclusion

The significance of diversity and inclusion was extensively recognized by interviewees, reflecting a collective understanding that diverse perspectives and inclusive environments are not just ethical imperatives but also crucial for the innovation and resilience of the field.

Education and mentorship emerged as fundamental elements to bolster diversity and inclusion, with suggestions further highlighting peer collaboration, funding support for inclusive initiatives, and the amplification of underrepresented voices. While participants outlined many specific thoughts on the importance of diversity and inclusion, some acknowledged it was not their area of expertise and did not offer specific suggestions.

Of all the questions asked, this question perhaps had the most varied responses. Some sharing it's the biggest hurdle to the field's advancement, while others skipped or disclosed they didn't have much to share. This variance in responses, from identifying diversity, equity, and inclusion (DEI) as a critical challenge to limited knowledge and understanding of the topic, underscores its complexity within the field. Below, we delineate the principal themes that emerged from these diverse perspectives.

"I think we have to make efforts, but time is the kind of force that is required to upend the structural inequalities, but also takes time for these things to happen. And so it needs intention. It needs positive action but also requires patience. And that's unfortunate." Participant 4

1. Education and Mentorship

Education and support throughout the entire learning process emerged as an important focus for improving diversity and inclusion within the token engineering field. Several participants acknowledged the importance of diversity and inclusion but expressed uncertainty about practical implementation. Some indicated a tendency to prioritize expertise over diversity in their hiring practices, revealing a potential diversity gap within the pipeline of available talent with the needed skills and experience. Participants acknowledged that cultivating a pipeline of educated talent for the field will require a multitude of educational efforts. Key challenges shared were a lack of availability of mentors, educational resources, and time, which were also mentioned in the "Challenges" and "Needs" sections of this report.

Participants shared that education is crucial for establishing and upholding standards in the token engineering practice. Simultaneously, they emphasized the importance of knowledge

sharing, specifically through mentorship, to enhance accessibility. Mentorship was said to transcend the mere transmission of skills and knowledge. It encompasses emotional support and the establishment of a nurturing relationship that can significantly impact an individual's journey in the field. This broader conception of mentorship recognizes the multifaceted challenges novices face, not only in grasping complex technical concepts but also in navigating the social and professional landscapes of the cryptocurrency and blockchain fields.

The time-intensive nature of effective mentorship was also acknowledged by participants, highlighting a notable tension between the desire to guide and support emerging talents and the practical constraints of professional commitments. This acknowledgment sheds light on the often overlooked aspect of mentorship – its demand on the mentor's resources, particularly time. Despite the recognition of its value, the capacity to engage in mentorship is limited by the existing pressures and workload faced by professionals.

“We need a foundational level. It would be great if kids at school could learn how to use wallets and have transactions. Educational programs. Cognitive learning process. Study groups, content production. We should have ambassadors going to universities and governments. Reaching the youth.” Shawn Anderson

2. Diverse Perspectives in Design and Decision-Making

Increasing diversity calls for actively involving varied perspectives in the design process. There is a difference between designing based on ideals of what certain groups need, versus asking about their needs and designing with them in mind, versus having those perspectives represented by the designers themselves inside the teams. The presence of underrepresented cultures, races, regions and genders in core roles within the development and thought process of web3 projects can create solutions that truly benefit the field and unlock global socio-economic development and coordination.

“Take a look at the list of cryptocurrency usage in the top 20. Do you have anybody from these countries on your team?” Success D. Ndu

While gender and race are commonly acknowledged aspects of diversity, Success D. Ndu highlighted the critical importance of incorporating individuals from diverse geographical and socio-economic backgrounds into teams. This perspective underscores the notion that inclusivity extends beyond mere representation to actively engaging and integrating voices from regions that are pivotal to the cryptocurrency market yet often underrepresented in development teams. By querying the absence of team members from countries that rank high in cryptocurrency usage, such as the Philippines, China, Nigeria, and Kenya, **the participant emphasizes the value of drawing on insights and experiences from those living in contexts where cryptocurrencies have significant traction and impact.**

This approach not only enriches the development and innovation process but also ensures that the technologies and systems created are truly global and responsive to a wide array of needs and challenges. The mention of a prominent blockchain's success in Africa serves as an illustrative example of how engaging with diverse communities can contribute to substantial progress and outreach, suggesting that a proactive stance in recruiting from a broad spectrum of societies is essential for fostering inclusivity and leveraging global potential in the cryptocurrency domain.

3. Funding

Participants pointed out that adequate financial support can significantly lower entry barriers for underrepresented groups, enabling more inclusive project initiatives and fostering a broader participation base. Pedro Parrachia noted the importance of funding initiatives committed to diversity and inclusion. According to them, while efforts can be made to educate and include, ultimately, it's the availability of funding that empowers their values and beliefs. This perspective underscores a shared understanding among respondents that targeted financial investment is essential for catalyzing meaningful DEI advancements in the field, by providing the necessary means for executing inclusive projects and amplifying marginalized voices.

Moreover, participants emphasized the value of connecting with existing legacy organizations and web2 initiatives as a strategy to enhance funding efforts. This approach is seen as a way to bridge gaps and leverage existing infrastructures for greater impact. The challenges of funding volatility and the difficulty in securing long-term research support were also highlighted, with Jessica Zartler noting, *"funding volatility greatly impacts the work,"* and Jeff Emmett adding, *"It's hard to receive funding for long-term research."*

4. Culture

Several participants highlighted that enduring challenges in DEI often stem from cultural and structural barriers. These include the intimidating complexity of the field, the predominance of regions like the US and Europe, and a pressing need for cultures within communities to be more welcoming and inclusive.

Building on this understanding, they articulated the significance of fostering personal growth, ensuring the creation of safe and supportive spaces, recognizing the time it may take for processes to evolve, and nurturing empathy within community interactions. Furthermore, establishing clear, respectful guidelines and practices that acknowledge and affirm individuals' pronouns and identities was identified as fundamental in advancing inclusivity within the web3 community, underlining a comprehensive approach to tackling these sociocultural challenges.

"I would say the main thing that's helped me is culture, and it's that people have actually been kind of nice and given me the time of the day even when I didn't know or understand what was

going on, or even now when I'm learning something. I think it's culture that people will come for, and stay for.” Kelsie Nabon

Summary: Education, mentorship, and financial support form the foundation for fostering diversity, equity, and inclusion in token engineering. Inclusivity goes beyond mere representation; opening core roles to individuals with diverse perspectives can benefit the entire field.

Thoughts on the Future

Participants were asked what they wished for the field and where they saw it headed in the next few years.

The responses were quite varied and often echoed sentiments already articulated in this report regarding addressing the challenges and needs within the field. When looking forward, many participants shared the imperative and potential of expanding experimentation. As Angela Kreitenweis mentioned:

“Tokens can be a tool to enter uncharted territory that we just can't cover with the currencies and how currencies are managed today. This potential would be and is still huge.”

Many participants expressed the viewpoint that the field is currently too narrowly focused, and even disappointment that more applications haven't yet been worked on. Danilo Lessa Bernardineli shared what many of our participants articulated:

“We have too many solutions for the same problem. At the same time, we have many problems in the world which token engineers for some reason, are not working on productively.”

While, reflecting on the future often conjured up a reiteration of the challenges and needs, nearly all participants saw an inevitable growth trajectory for both research and practice regarding token-based systems and that studying and sharing what has already been done is needed and helpful to the field's future formation.

Here are some of the key themes and messages from participants on what they expect or hope for the future:

Better Models, Standards, and Regulations

The participants shared a desire for more experimentation and expansion of the field. Going beyond what has already been done, while simultaneously identifying, creating, and elevating standards of what is already working well. Documenting prototypical practices does not need to

constrain the field, although it runs that risk, rather the sentiment was to develop a base layer of knowledge to advance foundational pieces like appropriate regulations and ethical standards, to increase transparency, accessibility, and even protections for those moving the work of the field forward.

- **Build better models**
 - *“It fascinates me still today, and has fascinated me in the past, that there’s this goal that I want from society as a whole...where individual goals and desires are compatible and in alignment with group goals and group desires. So where the individual need for something is not a cost for the group.”* Simon de la Rouviere
- **Create and elevate the standards**
 - *“If we fail to place standardization at the top of one of our priorities in the near future, we will find ourselves in an ever more, let’s say, unfortunate series of circumstances where we see projects that lose a lot of value and a lot of safety for individuals.”* Participant 30
- **Establish appropriate regulations**
 - *“And I think if you have standardization if you say hey, look, we’ve standardized this, and there’s actually an internal ability to say this is a good system and this is not a good system from these criteria, then you’re giving yourself the degree of freedom that a regulator will say they actually know what they’re talking about.”* Participant 30

Diverse Teams with a Multidisciplinary Approach

The participants depicted a future where transformative growth will come from embracing diversity in teams and methodologies. Global collaboration will benefit from building inclusive infrastructures and token engineering will evolve to leverage more multidisciplinary approaches focused on real-world application. Contributors will need to engage in deeper conversations and challenge assumptions, ultimately transcending the field’s current introspective tendencies and embracing its potential to serve broader societal needs.

- **Leverage the strength of diverse teams**
 - *“So if we want to build social, financial, economic, and governance infrastructures that are capable of serving communities that are all over the world, then you really gotta get engineers from all over the world. I think that this degree of professionalization is something to strive for.”* Michael Zargham
- **Lean heavily on a multidisciplinary approach**
 - *“That adoption of the system as a dynamic living entity is always going to require disciplines that you might not otherwise have thought are important.”* Participant 30
- **Be less insular and serve public-demanded work**
 - *“I think that right now the token engineering system as I perceive it, tends to have too much of a Dogecoin dynamic. It’s like right now it focuses so much on itself,*

rather than being of service to other fields and places.” Danilo Lessa Bernardineli

AI to Accelerate and Provenance to Differentiate

Participants shared visions of harnessing the power of AI and emphasizing provenance to drive innovation and differentiation. While AI is already being used within the field, participants articulated a future that will integrate AI more deeply to enhance the efficiency, scalability, and adaptability of systems. As a deeper history of on-chain data accumulates, the token engineering field is well-poised to contribute credible solutions to and integrate the use of provenance as systems evolve and the world of on-chain data and its applications for use grow. The use of AI and the ever-more-present need for provenance and reliable data will help the field evolve and address the increasingly complex and dynamic use of token and their systems.

- **Embrace AI to accelerate efforts**
 - *“Algorithmic controllers, AI leveraging monitoring analysis, and policy suggestions - much more adaptive systems. So right now there's this trepidation towards evolving your economics. But for the most part, EIP 1559 was just that. It was an evolution in the economics of Ethereum and has played a vital role in the sustainability of Ethereum and even people's projection of its value. So we shouldn't shy away from those kinds of more adaptive systems. So I think that will also be part of the future.” Participant 35*
- **Differentiate with applications of provenance**
 - *"Something that the entire space has kind of neglected, and that I'm starting to pay attention to more—which I think is very important, that needs to be surfaced more—is that over time, as this technology has existed, it has built up history and provenance about how things have happened, and we have this record of stuff that has happened, and that needs to, more and more over time, become the primary selling point of why this technology matters.” Simon de la Rouviere*

More Experimentation and Education

In the coming years, token engineering is set for radical experimentation and educational advancement to nurture systematic imagination and transcend traditional boundaries. Participants stressed the importance of interdisciplinary collaboration, individual exploration, and a deep engineering culture to foster innovation and trust. Education is deemed crucial, with calls for mentorship programs, structured materials, and increased mainstream recognition to elevate the field's accessibility and adoption. Participants desired the cultivation of a dynamic ecosystem that champions audacious experimentation, continuous learning, and inclusive and transparent access to resources.

- **Create more systematic imagination**

- *"I'm thinking many years in the future and the aspirations that I have for how the way in which emerging technologies can contribute to building, operating, maintaining and governing, sort of healthy infrastructures and institutions that maintain those infrastructures."* Michael Zargham
- **Audaciously experiment and be inspired, not confined by the past**
 - *"I really like Will Wright, who is the designer of The Sims and all these types of simulation games. Really great game designer. He says if you are obsessed about one particular thing sufficiently enough, you can make a game about it. You know, like if you look into a leaf or a stone long enough, you can create a game from that. So in a similar way, I think it depends on what obsessions are you bringing to the table and how can you utilize them?"* Cem Dagdelen
- **Educate and continue to elaborate**
 - *"Connecting and identifying talents, soon, and plugging them into the right mentorships in certain cases and economic contexts for practices in other cases, where they can actually get their hands dirty. That will be something extremely helpful. So something that links the exploration, pedagogy research dimension to the practice as sort of like parallel tracks. That will be next level."* Cem Dagdelen

Nearly all participants depicted a bright future for the field, with a few simultaneously emphasizing the biggest impediments to progress. Primarily a lack of regulatory clarity and that *"some of the deeper structural issues will only be addressed if we have sufficient funding and time to dig into those."* Jeff Emmett.

Wishes for the Field

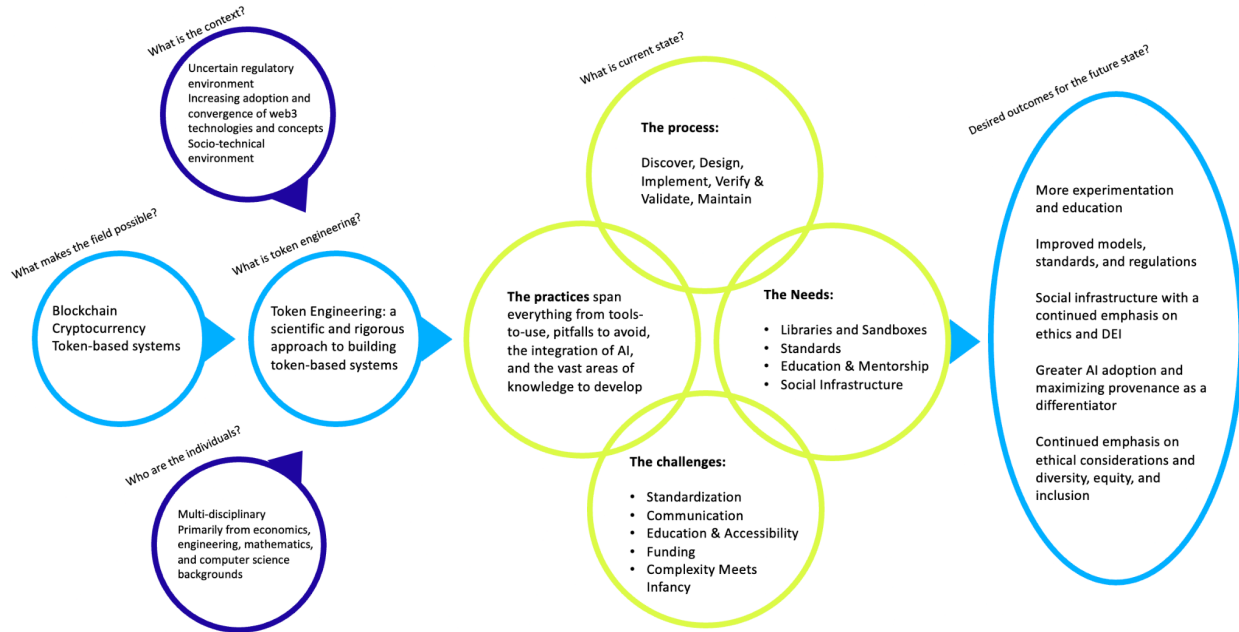
Participants expressed a variety of specific hopes, or wishes, for the future of token engineering. They envision a field that allows people agency over their participation in digital systems' design, governance, and ownership. There's a shared desire for increased participation, shared language, and knowledge exchange within the ecosystem, with an emphasis on responsible designing and engineering principles.

Many participants expressed aspirations for the field to evolve beyond mere financial tools, exploring tokens' broader meanings and applications. Ultimately, interviewees envision a token engineering landscape that promotes wealth-sharing, challenges traditional notions of currency creation, and encourages critical inquiry into societal values.

I wish _____.

- *“to see it be a very diverse, particularly geographically, obviously, gender and race as well, but I think geographically goes a really long way”*. - Michael Zargham
- *“For digital systems, that people have some agency over their ability to participate in the design and or ownership, governance, operation and maintenance of those systems.”* - Kelsie Naben
- *“More people to enter the ecosystem.”* - Participant 4
- *“many more projects better understand and also better understand how to integrate token engineering work in their projects. And I think this is rather a matter of time”* - Angela Kreitenweis
- *“more projects building using these principles of responsible designing and responsible engineering. Then we need more tools by the technical or educational infrastructural ones, and ways for people to be able to build and develop and test and release the token engineered products and services that they can conceive.”* - Participant 4
- *“My fantasy is to have an AI graduate student where I spend the same amount of time working with him than I would with a real graduate student. And then in two years, they have developed a highly specialized knowledge base.”* - Octopus
- *“Take on more meaning than just fungible currencies.”* - David Sisson
- *“We would make money which means that wealth is having enough to share”* - Andy Tudhope
- *“People, education, primarily, to look beyond token as a financial tool.”* - Participant 22
- *“To see more experiments with tokens but somehow we need to have an infrastructure that we can rely on”* - Ataberk Casur

Visual Summary of the Findings



Conclusion

Token engineering reveals a dynamic and multidisciplinary domain at the intersection of technology, economics, law, and ethics. And token engineers are the humans at work, blueprinting the token-based systems that are foundational to web3.

This study explored the question, “*What is token engineering?*”, and unearthed the definitional bounds, processes, practices, challenges, and needs of the current state of the field. Participants also articulated a view of the future and how they hope the field continues to advance.

The insights and reflections from participants were predominantly practical but inherently philosophical. Incentive mechanisms, governance structures, and the possibilities for building socio-technical complex systems have the potential to profoundly impact society, particularly in terms of human coordination and resource accessibility.

While there are many challenges intrinsic to the complexity and nascent nature of the field, participants articulated pathways for advancement. Particularly, they wished for more experimentation and education, along with establishing better models, standards, and regulations. The lightning pace of the industry underscored the need for diverse, multidisciplinary teams, AI adoption, and leaning into provenance as a distinct differentiator for this field. Participants articulated a hopeful future with many ways forward in an ever-expanding field.

The findings from this study highlight the importance of embracing the paradoxes and continuing to experiment, debate, and collaborate. Participating in the field requires working at the intersection of social sciences and technical knowledge while upholding standards of quality and ethical practices. The findings in this report provide a jumpstart for those interested in studying or participating in this field.

The token engineering domain is as intricate as it is transformative, as nebulous as it is promising. As the world adapts to the tectonic shifts triggered by decentralized technology, we hope the findings of this study serve as a solid foundation, resulting in more informed, ethical, and equitable token engineering practices. In these words and insights, we see the tools for reflection, and the wisdom to chart the course ahead.

"I do believe in continuous evolution. Sometimes it's going to be faster. Sometimes it's going to be slow. So I'm not concerned. The important thing is everyone is learning and making things in the open" Danilo Lessa Bernardineli

Discussion Guide

"I'm excited that this study is occurring and some of the questions that are being asked. I think deep down my wish for all of this stuff is that it kind of works in ways that are broadly positive and constructive. I think that's really important." Kelsie Nabon

The token engineering field is expected to undergo significant evolution. As blockchain technology matures and becomes more integrated into various industries, token engineering will likely see a surge in demand and innovation, alongside an increasing need to prove the legitimacy and trustworthiness of its solutions. We can anticipate a heightened emphasis on creating robust token systems that prioritize sustainability, fairness, and resilience.

Interdisciplinary collaborations will flourish, drawing expertise from fields like social science, mathematics, economics, computer science, and game theory to design sophisticated incentive structures, funding mechanisms, social rights components, and governance solutions. Deeper reflection and dialogue are required for the continued growth and refinement of token engineering as a pivotal discipline shaping the future of decentralized systems and digital economies. In this pursuit, questions either directly from the participants or from the analysis process were identified to prompt additional inquiry. The intent of this guide is to:

- 1) Acknowledge the nascency and evolution of the field:** token engineering is in a very early and malleable phase of determining the bounds and ideas to shape its future and therefore this guide encourages contemplations and conversations to grapple with the impact the current state of the field has on the future state.
- 2) Spark discussion with questions not answers:** disappointments expressed by participants were often from noticing how narrow the field has become in its use of tokens and therefore this guide emphasizes perspective expansion.
- 3) Advance the call for more participation and education:** many participants shared a need to connect with others and to engage in deeper conversations with people building in the space and therefore this guide can serve as one way to facilitate those efforts.

This study aims to support the legitimacy of these needs and encourage the readers to contribute to their areas of interest, advocating for a more holistic approach to system design that considers the long-term and profound impact this discipline is capable of having in society.

You can use these questions to reflect on your own or to discuss with others about the emerging field of token engineering:

- What makes token engineering a frontier field?
- What are the risks and rewards you associate with token engineering?
- What are the various uses of tokens? What uses are currently underrepresented?
- What are the primitives that compose the token engineering practice?
- How would you describe the end-to-end token engineering process and the roles needed to ensure its effective completion?
- What role do our assumptions play in the quality of token engineering outcomes?
- What are the implications of thinking of token engineering as a person or role versus a process that can't be performed by a person alone?
- Who is liable for the token engineering process?
- What does a successful token engineering outcome look like? How do we know? What criteria or metrics should be used to evaluate?
- When hiring someone to help build a token-based system, what criterion would you want to ensure they met?
- What is a token engineer ultimately responsible for?
- What are the 3 questions we hope every token engineer would be able to answer?
- What is needed to bolster the credibility and legitimacy of token engineering as a field and token engineers as professionals?
- What structures can help facilitate effective token engineering practices when formal education is not available, accessible, or desirable?
- What are the least obvious, yet relevant educational resources for token engineering?
- What are the ideal career paths or helpful career experiences for a token engineer?
- What bridges of coordination exist between groups building token-based systems?
- How can more avenues for collaboration be paved in ways that people can find, see, and use them?
- What are the implications of the prevalence of Ponzi schemes within web3 and the token engineering field?
- What role does ethics play in the blueprinting of token-based systems and their potentially exploitive or regenerative nature?
- In the interviews, ethical considerations were majorly framed by safety, manipulation, and accessibility concerns. How might ethical questions extend beyond these topics?
- What are the future implications for the token engineering field of present-day decisions on topics like regulation, ethics, diversity, artificial intelligence, compensation, standardization, and education?
- What are the hopes and fears you have for the future of the token engineering field?
- What other questions should we be asking?
- What is the most important question to ask yourself now?

Recommended Literature

We, the editors and team of this study, extend our sincere acknowledgments and gratitude to the resources that have influenced the field of token engineering, along with the individuals who have contributed to their creation.

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